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SUPERSEDING
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7 October 1983

MILITARY SPECIFICATION

SEMITRAILER, VAN: EXPANSIBLE, EXHIBITION, 12-TON, COMMERCIAL

This specification is approved for use by the U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers a 12-ton payload van type semitrailer with expanding side sections for housing and transporting exhibitions and audio-visual equipment. The semitrailer shall be equipped with tandem axles, an air ride suspension system, an air ride upper fifth wheel plate, air conditioning and heating systems and 120/240-volt electrical circuits. (See figures 1 through 12). Semitrailers procured under this specification are commercial items which shall be warranted by the manufacturer as specified in the acquisition documents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-2330

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SPECIFICATIONS

* FEDERAL

* A-A-393 - Extinguisher, Fire, Dry Chemical (Hand Portable).

* ZZ-T-381 - Tires, Pneumatic, Vehicular (Highway).

MILITARY

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.

MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance.

MIL-W-13518 - Wood Preservative, Tetrachlorophenol and Pentachlorophenol, Surface Sealing Compound.

STANDARDS

* FEDERAL

* FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.

FED-STD-595 - Colors.

MILITARY

MIL-STD-193 - Painting Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.

MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.

MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

MIL-STD-1473 - Standard General Requirements for Color and Marking of Army Materiel.

MIL-STD-1474 - Noise Limits for Army Materiel.

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.

Federal Motor Vehicle Safety Standards.

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(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the non-Government document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM Standards.
ASTM Designation: E380 - Metric Practice (DoD adopted).

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
NFPA 10 - Portable Fire Extinguisher.
NFPA 70 - National Electrical Code.

(Application for copies of NFPA publications should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SAE, Inc.
Standards and Recommended Practices.

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).

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J700	- Fifth Wheel Kingpin - Commercial Trailers and Semitrailers (DoD adopted).
J701	- Truck Tractor Semitrailer Interchange Coupling Dimensions.
J702	- Brake and Electrical Connection Locations - Truck-Tractor and Trailer (DoD adopted).
J844	- Nonmetallic Air Brake System Tubing (DoD adopted).
J875	- Trailer Axle Alignment.
J1292	- Automobile Truck, Truck-Tractor, Trailer and Motor Coach Wiring.
J1402	- Automotive Air Brake Hose and Hose Assemblies.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA
California Vehicle Code.

(Application for copies of the State of California Vehicle Code should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

(Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

UNDERWRITERS LABORATORIES, INC. (UL)
Standards

UL 490	- Standard for Attachments, Plugs and Receptacles.
UL 498	- Attachment Plugs and Receptacles.
UL 674	- Standard Electric Motors and Generators for Use in Hazardous Locations.

(Application for copies of UL publications should be addressed to the Underwriter's Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards),

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the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.19, the vehicle, components, assemblies and accessories to be delivered under the contract shall be standard or optional items which meet or exceed the requirements of this specification. All chassis items shall be as represented in the chassis manufacturer's technical data, and special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity prior to delivery of the vehicle. A first article shall be furnished.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Exterior treatment and painting. Treatment and painting shall be in accordance with MIL-STD-1223, except for the following:

(a) The exterior color shall be gloss white, matching Dupont "Dulux" white, number 93-21667. In MIL-STD-1223, paragraph 5.1.4 shall not apply;
(b) Underslung cabinets shall be gloss black, matching Dupont Imron 99U;
(c) Running gear, axles, chassis underside, gooseneck underside, landing legs, jacks, suspension (except the air bags and brake components) shall be finish painted gloss black, matching Dupont Imron 99U;
(d) Stairs and handrails shall not be painted, except the standing surface of the stairways shall be treated as specified in 3.4.3.9.1, with the color being yellow.

* 3.1.1.1.1 Interior treatment and painting. The vehicle interior shall be treated and painted in accordance with MIL-STD-1223, except as follows:

(a) The ceiling and walls of the projection booth shall be flat black, matching Dupont Imron 99U;

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(b) The ceiling in the service and theater areas shall be semigloss light brown, matching Dupont enamel color chip No. 29607;
(c) The plywood walls in the service and theater areas shall be finished with a semigloss clear varnish (see 3.1.1.3);
(d) The topside of the plywood subfloor, the topside of each expansible section plywood floor and the plywood folding floor sections on all sides shall be coated with a clear protective finish (see 3.1.1.3);
(e) The exposed exterior of switch boxes, junction boxes, electrical controls, circuit breaker boxes and conduit shall be gloss blue, in accordance with table III of MIL-STD-1473. Gloss blue shall match color chip No. 15102 of FED-STD-595;
(f) The interior of electrical areas, such as switch boxes, junction boxes and circuit breaker boxes, shall be gloss orange matching color chip No. 12246 of FED-STD-595, in accordance with MIL-STD-1473;
(g) Air ducts, if exposed, shall match the color of adjacent surfaces;
(h) The air conditioner(s) shall be treated and painted in accordance with 3.9.19;
(i) The interior color of the stowage compartments shall match the vehicle exterior color (white).

* 3.1.1.2 Markings. Identification markings and data plates shall conform to the Army requirements of MIL-STD-1223.

3.1.1.2.1 Instruction and identification plates. Plates indicating tire pressure; fill level of liquids; operating instructions for and source of the expansible side components; operating and servicing instructions for the electrical systems; operating and servicing instructions for the rear air suspension system; operating and servicing instructions for the upper fifth wheel air suspension system; operating and servicing instructions for the air conditioning and heating systems; schematics of all valves, valve controls, cylinder equipment; complete operating instructions for the expansible side sections; and miscellaneous plates specified herein shall be securely fastened in protected but visible locations. Each control shall be marked with a plate depicting the function of that control. Individual markings shall be adjacent to or on each control.

* 3.1.1.2.2 Plates. Plates shall conform to composition A (class 1 or 2) or composition C of MIL-P-514 and shall be affixed to the vehicle in close proximity to the items to which they relate. Plates shall be sized as required. The color of the plates shall conform to table I of MIL-P-514. The Government will not furnish any plates as specified in

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paragraph 3.7 of MIL-P-514. Foil shall not be employed on the vehicle exterior or on plastic surfaces.

3.1.1.2.3 Semitrailer identification plate. An identification plate shall be affixed inside the lower front corner of the service area door. The plate shall conform to 3.1.1.2.2, and shall be approximately 10 inches wide by 8 inches high or as required to show at least the following information:

IDENTIFICATION PLATE
SEMITRAILER, VAN: EXPANSIBLE, EXHIBITION,
12-TON, COMMERCIAL

MANUFACTURER _____	MODEL NO. _____
NSN _____	MILITARY SPEC. NO. _____
REGISTRATION NO. _____	SERIAL NO., MFG. _____
CONTRACT NO. _____	VAN WIDTH, OPEN _____
LENGTH _____	VAN WIDTH, CLOSED _____
HEIGHT _____	CURB WT. KINGPIN _____
CURB WT. REAR BOGIE _____	CURB WT. LANDING LEGS _____
PAYLOAD:	TOTAL CURB WT. _____
CENTER OF GRAVITY W/O PAYLOAD	GROSS WT. _____
ABOVE GROUND _____	DATE OF MFG. _____
FORWARD OF BOGIE _____	U. S. PROPERTY _____

3.1.1.3 Wood treatment. All wood shall be treated in accordance with MIL-STD-1223.

3.1.1.4 Masonite. When masonite is used in the construction between the exterior panels and the roof bows and the interior panels and the vertical posts or crossmembers, the material shall be tempered and shall be treated in accordance with type I, class I of MIL-W-13518.

3.1.1.5 Dissimilar metals. When parts of dissimilar metals are in contact, the contact faces shall be primed with a highly resistant alkaline primer to prevent galvanic and electrolytic action. Materials used shall be compatible with each other in accordance with MIL-STD-193.

* 3.1.1.6 Rustproofing. The vehicle shall be rustproofed in accordance with FED-STD-297. In addition, the following shall apply:

(a) Metal surfaces shall be cleaned, treated, primed and painted prior to rustproofing. All surfaces to be rustproofed shall be primed with a highly resistant alkaline primer;
(b) All the enclosed areas, including under the fifth wheel upper plate, but not to interfere with the hinge areas of the plate, shall be rustproofed;

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(c) Boxed-in areas, such as light cavities over the conduit, clips and brake line metal tubing and structural members shall be rustproofed;
(d) All exposed areas under the vehicle, including stowage compartments and exposed areas of the floor pan, except the brake, electrical and air suspension systems shall be rustproofed;
(e) There shall be no rustproofing compound on the vehicle visible from an eye level height of 3 feet or more to the sides, front, or rear of the semitrailer.

* 3.1.1.7 Stowed equipment. Equipment to be located in the stowage compartments shall be provided with tiedowns or other means of securing the equipment for travel.

* 3.1.1.8 Fire extinguishers. Four fire extinguishers shall be installed inside the vehicle. The fire extinguishers shall be in accordance with National Fire Protection Association No. 10. Two of the extinguishers shall be 20-pound, 1211 Halon type. The other two shall conform to type II, class 1, size 20 of A-A-393. Fire extinguishers shall be mounted on brackets with cam type clamps and shall be readily accessible. The fire extinguishers shall be located as follows:

(a) One of each type in the service area, accessible through the service area door;
(b) One of each type in the projection booth, accessible through the projection booth door.

3.1.1.9 Flame resistance. Flame resistance shall be in accordance with Federal Motor Vehicle Safety Standard No. 302.

3.1.1.10 First aid kit. A first aid kit shall be installed in the service area in an accessible location. The first aid kit shall be in accordance with Federal Motor Carrier Safety Regulation 396.96, type B commercial type kit. The first aid kit shall be constructed as specified in Federal Motor Carrier Safety Regulation 396.96 (1) and (2).

3.1.1.11 Hydraulic fluid. Hydraulic fluid, if used, shall be in accordance with MIL-H-5606 or equivalent. The flash point shall be not less than 180 degrees Fahrenheit (°F) and shall allow hydraulic component operation in a ambient temperature of minus 40°F.

* 3.1.1.12 Lubricants. Commercial lubricants to be used shall be determined by the contractor and depicted on a plate installed in a readily accessible location in the service area. The plate shall indicate ambient temperature range for the various lubricants. The plate shall conform to 3.1.1.2.2.

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3.1.1.13 Electrical entrance receptacle warning plate(s). Warning plate(s) shall be secured to the exterior in the area of the electrical entrance receptacles. Plate(s) shall be at least 5 inches by 3 inches in size and shall indicate the voltage, phase, Hertz (Hz), number of wires and the ampere capacity. The plate(s) shall conform to 3.1.1.2.2.

* 3.1.1.14 Rear wheel splash and stone throw protection. Rear wheels shall have rubber mud flaps. Splash and stone throw protection shall be in accordance with SAE J682. A metal strip not less than 1/8 inch thick and not less than 1 inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or the bolt nuts from damaging the mud flap. The mud flaps shall be readily removable with common hand tools.

* 3.1.1.15 Temperature extremes. The vehicle, vehicle components and materials shall be capable of operation and use in ambient temperatures from minus 20°F to plus 125°F without evidence of deformation, cracking, or unusual wear. The temperature extremes shall not affect the vehicle operation and the warranty, expressed elsewhere in procurement documents.

3.1.1.16 Screw thread and fittings. Screw threads and fittings shall be based on United States Standards except as specified in 3.1.1.16.1.

3.1.1.16.1 Metric components in equipment. Metric components in equipment shall be identified. The contractor shall identify metric tool requirements for all types of maintenance. Metric usage shall conform to ASTM E380. Metric equivalents may be used either as a primary standard of measurement, or as a secondary standard. Where metric equivalents in parentheses are not considered practicable, a tabulation of equivalents shall be depicted.

* 3.1.1.17 Pocket for manual and equipment record folder. A pocket for a vehicle operator's manual and a Government Log Book Binder or equipment record folder (NSN 7510-01-065-0166, 8-1/2x10x3 inches in size, to be installed by the user after receipt of the vehicle) shall be installed in an accessible location in the service area. Equipment Control Record DD Form 2408.9 will be inserted in the Government Log Book Binder or equipment record folder.

3.1.1.18 Radio interference suppression. The 12-volt electrical system shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

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* 3.1.1.19 Electromagnetic interference. Electromagnetic interference suppression shall be furnished for at least the following 120-volt and 240-volt components:

(a) Electric motors
(b) Heaters
(c) Air conditioner(s)
(d) Vehicle interior lighting system.

3.2 General design.

* 3.2.1 Federal Motor Vehicle Safety Standards and Regulations. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

* 3.2.2 Interior sound level. The interior sound level in each of the three semitrailer areas, with the air conditioning system operating and then with the heating system operating, shall comply with category F, table I of MIL-STD-1474 and shall not exceed 65 dB(A).

3.2.3 Curb weight. The curb weight of the semitrailer shall include the weight of the completed semitrailer, with all attachments, accessories, installed equipment and stowed equipment.

* 3.2.4 Payloads. The rated payload capacity for mobile operations shall be not less than 24,000 pounds. The rated payload capacity with the sides expanded shall be not less than 9,000 pounds.

* 3.2.4.1 Payload distribution, mobile operations. To determine conformance to the mobile requirements specified herein and for tests, the rated mobile payload shall be evenly distributed in the semitrailer with the semitrailer's expansible side sections retracted and the semitrailer in travel mode.

* 3.2.4.2 Payload distribution with side sections expanded. The rated payload capacity with the side sections expanded shall be not less than 9,000 pounds, concentrated in 30 individual, 300-pound per square foot units, located singly or grouped together at any location(s), at the Government's option; except not more than 4,500 pounds, or 15 individual, 300-pound per square foot units, need be located on any one single expansible side section at any one time.

3.2.5 Gross weight. The gross weight of the semitrailer shall consist of the curb weight plus a payload of 24,000 pounds, distributed as specified in 3.2.4.1.

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3.2.5.1 Gross weight on the upper fifth wheel plate. The gross weight on the upper fifth wheel plate, with the payload distributed as specified in 3.2.4.1, shall not exceed 45 percent of the gross weight of the semitrailer.

3.2.6 Ratings. Ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be made available to the engineering office of the procuring activity for review and acceptance or rejection.

* 3.2.7 Dimensions. Semitrailer dimensions and clearances, without payload, uncoupled from the towing vehicle, resting on the landing legs, on level ground, with the vehicle level, with the expansible sides retracted in travel mode and then with the sides expanded in exhibition mode, shall be as specified in table I.

TABLE I. Dimensions.

Overall exterior:	
Length (plus 0 inches, minus 3 inches)	480
Width, exclusive of rub rails, (plus 0 inches, minus 1 inch)	96
Height, maximum (inches)	156
Upper fifth wheel height (distance from ground to underside of the air ride upper fifth wheel plate at the centerline of the kingpin) (plus 1 inch, minus 0 inches)	48
Kingpin to front of semitrailer (plus 1 inch, minus 0 inches)	18
Swing radius (plus or minus 1 inch)	51
Landing wheel clearance (distance from centerline of the kingpin to the leading edge of any point on the vehicle, 6 inches or more below the upper fifth wheel plate) not less than (inches)	90
Ground clearance:	
Under the axles and slack adjusters, not less than (inches)	14
Under the front landing gear assembly, retracted (inches)	14
* Under the rear landing gear assembly, retracted (inches)	See 3.4.7
Under the stowage compartments (and providing 15-degree approach angle measured from the centerline of the kingpin to the bottom forward edge of the stowage compartments) not less than (inches)	14
Ceiling height, nonexpansible area, approximately (inches)	90
Service area, exterior:	
Overall length, measured from exterior front of the semitrailer to the expanded front wall (plus or minus 2 inches)	125

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TABLE I. Dimensions. - Continued

Expansible sides, exterior:	
* Overall length, maximum (inches)	280
Overall width, sides expanded (plus 4 inches, minus 2 inches)	170
Ceiling height, approximately (inches)	86
Theatre area, sides expanded, interior dimensions:	
Length, not less than (inches)	274
Width, not less than (inches)	165
Height, not less than (inches)	86
Projection room:	
Interior length, not less than (inches)	60

* 3.2.8 Electrical systems. All 12-volt electrical systems shall conform to SAE J1292. All 120- and 240-volt systems shall conform to NFPA No. 70, the National Electrical Code. All circuits in all semitrailers under contract shall be wired and identified identically.

* 3.3 Performance. Performance requirements for the semitrailer shall be met when fully equipped, completely serviced, and for mobile operations coupled to a truck tractor and towed over improved and unimproved, hard, uneven terrain. The semitrailer shall show no evidence of failure or malfunction of the vehicle components, attachments, or accessories. The expansible sides, interior floors, landing gear assemblies, and all cabinet and personnel doors shall be exercised as required to verify that no malfunction exists and that road dust will not accumulate in the interior.

* 3.3.1 Towing speed and trailing ability. The semitrailer shall follow a truck tractor at sustained speeds of at least 58 miles per hour (mph) over dry, level, smooth, prepared hard surfaced roads, and of at least 35 mph over gravel roads. The vehicle shall follow the truck tractor without exceeding the tracking deviations specified in Federal Motor Carrier Safety Regulation 393.70(a).

3.3.2 Turning ability. The semitrailer, when coupled to a truck tractor, shall be capable of assuming an angle of 90 degrees to the truck tractor without damage to the semitrailer or to the truck tractor. Turning ability shall be demonstrated when the semitrailer is coupled to a 96-inch wide truck tractor having a landing wheel clearance dimension of 90 inches, in accordance with SAE J701 for truck tractors with tandem rear axles.

* 3.3.3 Grades and slopes. The semitrailer, coupled to the truck tractor, shall be capable of operating on 0 to 20 percent grades and on 0 to 12 percent side slopes, up and down the respective grades and

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slopes, without any interference or malfunction. When the semitrailer is on any incline specified, the semitrailer shall not touch any component of the truck tractor, except where the two are in normal contact in normal use.

3.3.4 Service brakes. The service brakes and the stopping distance of the semitrailer shall be in accordance with Federal Motor Vehicle Safety Standard No. 121.

3.3.4.1 Emergency brake. An emergency brake shall be furnished. The emergency brake shall be in accordance with Federal Motor Vehicle Safety Standard No. 121.

* 3.3.5 Waterproofness. The semitrailer, including the stowage compartments, shall be waterproof to prevent the entrance of water when exposed to a spray of 40 pounds per square inch gage (psig) pressure at the entrance to each nozzle, located two feet from the vehicle, for at least 30 minutes. Nozzles shall be not more than 5 feet apart and shall provide full water coverage. When the expansible sides are retracted, water spray shall be projected on all sides, on the bottom and on the roof. When the expansible sides are in the expanded position, the water spray shall be projected on one side and the roof of the vehicle, and then on the opposite side and roof of the vehicle. Under these conditions, the gaskets, seals, rivets and exterior lapped panels shall show no evidence of water leaks, except at the vertical shaft gearing at each corner of the expansible section. Means shall be provided to drain such water to the ground. Permissible weepage shall be tear drops of water in not more than three areas per side. Waterproofness shall be determined both before and after interior plywood wall panels and wall and ceiling insulation are installed, after expansion and loading of the expansible sides and after road testing.

3.3.6 Airtightness. The semitrailer, including the stowage compartment, shall be airtight during transit to prevent dust from entering the interior.

* 3.3.7 Environmental control system. An environmental control system shall control the environment within the semitrailer when the sides are expanded, all doors are closed and the semitrailer is in ambient air conditions ranging from minus 20°F with a wind of 15 mph to plus 105°F dry bulb with a wind of 7-1/2 mph. With the cooling unit operating, the specified inside temperature shall be maintained with internal heat from lights, evaporator fan motor, and the heat gain with 16 personnel within the semitrailer. With the heating unit operating, the specified inside temperature shall be maintained with no internal heat gain from any other source. The environmental control system shall

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maintain, at the thermostat, a temperature of 73.4° (23°C) dry bulb, plus or minus 5°F (2.8°C) and 50 percent maximum relative humidity, and shall control the temperature throughout the semitrailer at a level of 5 feet above the finished floor at 73.4°F (23°C) plus or minus 5°F (2.8°C). Controls shall enable functions as specified in 3.9.3 through 3.9.9.

* 3.3.8 Shock and vibration. The entire finished and completed semitrailer shall withstand the road conditions specified. Semitrailer components, fasteners and accessories shall show no loosening, damage, or malfunction when loaded as specified in 3.2.4 through 3.2.5 and when operated as specified in 3.3 through 3.3.4.1.

* 3.3.9 Deflection. With the side sections expanded and the semitrailer loaded as specified in 3.2.4.2, with the 300-pound per square foot loads shifted into all possible combinations of distribution and concentration (e.g., 4,500 pounds grouped on any extreme corner of one expansible section, 4,500 pounds lined up along any one expansible outer wall, 9,000 pounds scattered throughout the semitrailer, etc.), the entire floor shall remain level at all points, within the measurable limits of a 1-foot and a 3-foot carpenter's level. The floor shall be capable of withstanding the concentrated loads represented by not less than 40 people, each weighing not less than 200 pounds, sitting in folding chairs having four legs; including loads imposed by a 200-pound chair occupant balancing back on one or two legs of the chair. No visible warpage or deflection between floor sections shall occur. After loading, the expansible sides shall retract and expand properly with no adjustments, off-tracking, binding or unusual efforts.

* 3.3.10 Service life. The semitrailer expansible side sections shall be capable of daily cycles of expansion, loading (see 3.3.9), and retraction for a period of not less than 1 year without ever untracking or binding and with no wear or tear of the seals sufficient to result in failure of the waterproofness requirements of 3.3.5. Leveling of the semitrailer required to meet these performance requirements shall be limited to the capabilities of the landing gear the contractor furnishes, measured within only the accuracy of the spirit levels the contractor furnishes.

* 3.3.11 Lubrication. All parts of the semitrailer which require lubricants, whether considered by the manufacture to be permanently lubricated or not, shall be provided with ready access for lubrication. Access panels or doors shall be installed as necessary, with hinged access door(s) used wherever space permits. All parts of the semitrailer chassis running gear, body and installed equipment which require lubricants shall be capable of being lubricated by one man in

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not more than 2 hours, including time to clean lubrication fittings and open and reclose access panels or doors. Any special lubrication equipment or lube gun extensions required to meet these requirements shall be provided with the vehicle and securely stowed in the stowage compartments.

* 3.3.12 Seals. Seals for the expansible sections shall have the service life specified in 3.3.10. All such seals shall be readily available, standard commercial products. All such seals, in both side sections, shall be capable of being removed and replaced with new seals by a two man crew in not more than 2 hours, without lifting equipment or metal cutting tools.

* 3.3.13 Folding floor. The folding floor shall be capable of being unfolded and then refolded and secured for transit, each operation in not more than 15 minutes, by two men, without more than 40 pounds of effort per man. Recessed hand holds, an electric motor driven hydraulic system or other means, all of the contractor's option, shall be provided as required to meet performance requirements, except winches shall not be used. No part of the system shall encroach upon the space in the theatre area or shall be visible from the theatre area, with the side sections expanded and the floor unfolded. The system shall conform to all safety provisions and regulations, human engineering factors and other considerations as applicable to the system chosen by the contractor. Appropriate caution plates conforming to 3.1.1.2.2 shall be provided.

3.4 Design and construction. The semitrailer shall be a fully enclosed van type body with expanding side sections. The semitrailer may have a level platform from front to rear, or a drop type platform, level forward and rearward of the gooseneck, all without wheelhousings. The semitrailer shall be equipped with air conditioning and heating systems, 120/240-volt electrical circuits, double front doors to service the air conditioning system, a single rear door and side and rear doors in the expansible sections, all as specified herein. The semitrailer, when in a static position, shall be supported by four adjustable, two-speed landing gear assemblies. The semitrailer shall be designed and manufactured to the highest standards in the industry.

* 3.4.1 Body. The semitrailer shall be of the body-on-frame type, with full length frame rails supporting the entire structure and the payload. No part of the body walls or roof shall be considered as contributing to the load-carrying ability of the semitrailer. Stressed skin type construction shall not be furnished. The semitrailer shall be designed and constructed from concept as an expansible van and shall not be a modified or redesigned conventional van semitrailer.

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* 3.4.1.1 Body framing structure. Framing for the body, underbody and roof shall be constructed of steel sections. Steel structural members shall be arc welded and reinforced at joints to fulfill the dynamic, static and expansible side sections' loading requirements specified herein. Where splices are necessary, they shall be designed to avoid stress concentration. Splice welds shall be continuous. Welding material shall be compatible with the material being welded. Structural members shall be designed to accept the loads imposed by the factors specified. Tire clearance shall be not less than 8 inches, without the use of wheelwells.

* 3.4.1.2 Floor pan. A complete floor pan shall be provided under the vehicle. The pan shall be not less than 0.050 inch thick, of aluminum alloy sheet. The floor pan shall be installed across the bottom of the semitrailer. The floor pan shall be watertight to prevent contamination of the insulation cavity so formed. The area of the floor pan inside the stowage compartment shall be painted white to match the exterior of the semitrailer. The floor pan shall be provided with removable covers for adjustment and servicing of the gear reducers. The area above the floor pan shall be insulated with material conforming to 3.4.1.13, with insulation not less than 2 inches thick.

* 3.4.1.3 Fixed subfloor. A fixed subfloor shall be provided, the full length and width of the nonexpansible section of the semitrailer (except see 3.5.5.5). The floor shall be of marine grade plywood not less than 3/4 inch thick (see 3.1.1.3). The floor shall be secured with self-tapping, corrosion resistant machine screws.

* 3.4.1.4 Folding floor construction. The floor of the nonexpansible area shall consist of four sections, each hinged with a stainless steel continuous hinge. Each of the sections shall be capable of folding toward the center of the main body and of being lowered to form a floor for the nonexpansible area. When the floor sections are lowered onto the fixed subfloor, a draftproof and weather-resistant theatre area shall result. The hinge area shall be weather-resistant and draftproof. Floor sections shall be on the same plane as the expansible floor. The floor shall be backed by suitable aluminum structural subframing to prevent the plywood from ever buckling, warping, splitting or cracking. Means of securing the folded floor against shifting, vibration and shock during over-the-road operation shall be provided. For mobile operation demonstration and tests, 1/2-inch thick carpeting or other suitable material shall be provided by the contractor and temporarily installed over the floor. The folding floor shall conform to 3.1.1.3 and 3.3.13. See 3.4.9 and 3.5.3.

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3.4.1.5 Tapping plates. Two steel tapping plates, parallel to each other and installed between the vertical members of both the curbside and streetside expansible walls, shall be furnished (see figure 6). The tapping plates shall be suitable for use in suspending and securing pictures, lighting fixtures and book racks. The two plates in each wall shall be of at least 11 gage (0.1196 inch) steel by at least 3 inches plus 0, minus 1/8 inch in width, with a space between the plates of approximately 29 inches. The top horizontal tapping plates shall be located approximately 11 inches from the interior of the finished ceiling to the top edge of the tapping plates. Tapping plate locations shall be marked on the interior plywood walls for use by personnel when drilling and tapping into the plates.

3.4.1.6 Framing structure. Body framing structure shall include at least the following: The streetside and curbside outermost longitudinal members at the roof level shall extend the full length of the vehicle and shall be utilized to join full width roof bows spaced on not more than 24-inch centers. Vertical members, consisting of corner and intermediate posts, shall be inserted between and employed to combine the roof structure to the chassis frame. The corner posts shall be one-piece, without splice, between the roof structure and chassis frame and shall be such as to present rounded corners on the vehicle. The intermediate posts shall be continuous without splice between the roof structure and the chassis frame except at the door opening areas, and shall be spaced on not more than 24-inch centers except different intervals will be permitted at door opening regions. Diagonal horizontal members between the vertical posts may be used to add to the rigidity of the truss. Horizontal members shall be provided between the vertical posts for the tapping plates specified in 3.4.1.5.

3.4.1.7 Rub rails. Not less than 11 gage (0.1196 inch) rub rails shall be provided along the streetside and curbside of the semitrailer. The rub rails shall be removable for replacement without damage to the body panels. The rub rails shall be sealed.

3.4.1.8 Drip mouldings. Drip mouldings shall be provided over all body exterior openings. Drip mouldings shall be sealed at adjoining surfaces. Rain water shall not drip from the expansible roof over any door opening.

* 3.4.1.9 Framing for expansible side section openings. Steel framing shall be provided around the perimeter of the open sections on each side of the semitrailer for the expansible side sections. The horizontal and vertical framing that envelopes the body side panels shall be sealed during installation. Corners shall be mitered and joints shall be tangent and sealed. The vertical and horizontal framing

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shall have sufficient development to compress against the gaskets or seals when the expansible side sections are retracted and when the side sections are expanded. The edges of the metal framing in contact with the gaskets and seals shall be blended to prevent the metal framing from deforming or cutting into the gaskets and seals. The metal framing shall be bolted to the sides.

* 3.4.1.10 Exterior body panels. Exterior body side, door, front and rear end panels and expansible side section side, end and bottom panels shall be aluminum alloy not less than 0.050 inch in thickness. Stretcher leveled, flat, noncorrugated panels with interior posts shall be furnished. The panels shall have not less than a 2-inch lap, and shall be sealed with a water-insoluble sealer. Side panels on the nonexpansible portion of semitrailer and side, end and bottom panels on the expansible sections shall be riveted on not more than 2-inch centers. Panels shall be free of any defects or imperfections that would affect the performance or the appearance of the semitrailer.

* 3.4.1.11 Roof panels. The roof panels of both the nonexpansible and the expansible side sections each shall be of one-piece, stretcher leveled, aluminum alloy not less than 0.040 inch in thickness. The roof panels shall be sealed with water-insoluble sealer and secured around the perimeter on not more than 2-inch centers. The roofs shall be crowned to permit drainage. The roofs shall be capable of being walked on by 200-pound personnel, clearing up to 6 inches of wet snow, without damage, prior to retracting the sides. Roof panels shall be secured to prevent panel vibration. Expansible side sections shall be provided with roof heaters below the roof panels to melt ice and snow (see 3.6.7).

* 3.4.1.12 Body attachments. The body shall be fastened together in a manner which will preclude loosening of bolts, screws and rivets, and cracking of welded joints. Fasteners shall be compatible with the material being secured. Fasteners shall be sealed during installation. Self-tapping screws shall not be used in areas subjected to stress. Wood shall not be used for load-supporting structural framing, except as shown in figure 9 for the rear partition. Body panels shall be properly riveted. All bolted and riveted attachments shall be sealed to prevent entrance of wind and water. The interior seams shall be coated with water-insoluble sealant. All exterior body framing drive screws, rivets and bolts shall be squarely installed and shall not be overtorqued. Stripped threads are not acceptable.

* 3.4.1.13 Insulation. Body insulation having a thermal conductivity ("K" factor) of not more than 0.26 British thermal units (Btu) per hour foot degree Fahrenheit, and a density of not less than

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1.50 pounds per cubic foot shall be installed to completely fill the cavities in the walls, ceilings, and exterior doors. Insulation shall be provided in the areas above the floor pans (below the fixed subfloor and below the floor of the expansible sides). Insulation in these areas shall be not less than 2 inches thick. The insulation shall not support combustion or sustain flame, mold, rot, or vermin, corrode metals, absorb odors, and shall be essentially odorless. The insulation shall not disintegrate under vibration or settle and leave voids. Insulation material shall be homogeneous and uniform in density and thermal efficiency. Insulation material shall be capable of flexing more than its containing structure without tendency to crack or rupture. Insulation shall be of the foam or fiberglass blanket type. The fiberglass blanket type insulation shall be fastened by the use of chemical bonding, adhesive tape, and corrosion-resistant clips to prevent sagging. The fiberglass blanket type insulation shall have a plastic or aluminum foil type vapor barrier on the interior side. Condensation in the insulation cavities shall not exceed 0.50 pound per hour. Means shall be provided to drain condensation to the ground but shall not allow road splash to enter the insulation cavities.

3.4.1.14 Materials. All materials shall be free of any defects or imperfections that may affect the performance and first quality appearance of the vehicle.

3.4.1.15 Sealer for waterproofing. Sealer shall be water insoluble and shall not harden. The sealer shall be capable of resisting weepage between any attachments where the sealer is employed. The sealer shall be capable of passing the water test specified herein. Sealer shall withstand the temperature extremes specified in 3.1.1.16.

* 3.4.1.16 Seals. The seals shall prevent the entrance of water, snow and dust into the semitrailer interior when in travel mode with the side sections retracted and when in exhibition mode with the side sections expanded. The seals shall be accurately fitted. The seals shall have flexural and compressability characteristics and shall not absorb moisture. Seals shall not project beyond the 96-inch width of the semitrailer. Seals shall not be painted. Seals shall conform to 3.1.1.15 and 3.3.12.

* 3.4.1.17 Rear end protection. Semitrailer rear end protection shall be in accordance with Federal Motor Carrier Safety Regulation 393.86.

3.4.2 Design load and stresses. The design load shall be equal to the product of 1.5 times the algebraic sum of the vehicle curb weight minus the unsprung weight of the suspension system plus the rated design

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payload uniformly distributed over the load area as specified in 3.2.4.1. The 1.5 factors shall include and adjust for the effect of a shock load (0.50) in stress calculations. When subjected to the computed design load, the normal stress in the load-carrying members shall not exceed 50 percent of the material yield strength. When the column loading is the critical loading condition, the column loading shall be computed in accordance with the latest edition of the American Institute of Steel Construction, Incorporated (AISC).

* 3.4.2.1 Stress analysis. A complete stress analysis for the vehicle shall be performed. The stress analysis shall include at least the following calculations and diagrams, based on the design load and loading when expanded as specified in 3.3.9:

(a) A diagram of the total sprung weight distributed on the kingpin and the axles. The sprung load distribution diagram, expressed in terms of an equation:
Design load = 1.5 (gross weight minus unsprung weight);
(b) The vertical shear diagram for the length of the vehicle;
(c) The bending moment diagram for the length of the vehicle;
(d) The loads, sections, shear and stresses in members subjected to maximum shear and moment, to include the analysis of the stress in attaching weldments or other fastenings;
(e) The loads, sections, shear, and bending moment diagrams at the kingpin and gooseneck area (upper fifth-wheel plate) members and the chassis understructure frame members including suspension area subframe members;
(f) The load, section, shear, and bending moment diagram for the most critical crossmember, to include an analysis of the stresses in attaching weldments or other fastenings;
(g) The loads, sections, stresses and maximum deflections in members subjected to maximum compressive forces;
(h) An analysis of the loads, sections, shear and stresses in the members of the perimeter framing specified in 3.4.1.9, with the design load and loaded under static conditions as specified in 3.3.9;
(i) The analysis shall show required strength in the semitrailer frame, subframe and floor to support the design load with the semitrailer roof and walls not considered as contributing to the strength of the semitrailer;
(j) An analysis of the loads, sections, shear and stresses in the entire expansible side box sections (see 3.4.9) when expanded and loaded as specified in 3.3.9, and empty during expansion and retraction. Wall and ceiling plywood paneling in the expansible sections shall not be considered as contributing to expansible side section strength, to insure loading, expansion and retraction operations prior to panel installation can be accomplished;

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(k) An analysis of each of the 3 roof sections showing the ability to withstand the loads imposed by 6 inches of wet snow;
(l) Calculations showing the body walls and roof can withstand a 50 mile per hour wind velocity;
(m) A list of all the assumptions and references utilized to provide the required data;
(n) The contractor's format for the calculations will be acceptable provided that the calculations are presented in a logical and straight-forward manner complete enough so that a reviewer can follow through the work, step-by-step without difficulty.

* 3.4.3 Doors.

* 3.4.3.1 Service area side door (gooseneck). A door to the service area, located on the curbside in the gooseneck area, shall be furnished. The gooseneck side door shall be located just forward of the expansible side section. The door shall not interfere with the construction of the expansible side section. The door opening shall be not less than 30 inches in width and not less than 78 inches in height. The door shall open toward the front of the vehicle.

* 3.4.3.1.1 Door construction. The door shall be hung to permit the door to fold back against the body side. Means shall be provided to retain the door in fully open position. The retaining device shall include a retaining spring and a door closure. The door shall be insulated. Door clearance shall not exceed 3/8 inch. The door shall be hung between box posts with heavy duty piano type hinges. The door shall be provided with bright stainless steel handles, latches and locks with key-operated tumblers. Door handles, latches and locks shall be recessed in the body panels. The doors shall be capable of being unlocked, opened and locked from inside and outside the vehicle. Handles, latches and locks selected by the contractor shall be subject to pre-approval/rejection by the Government. The door shall be provided with a tapered door sill.

3.4.3.2 Theater area side door (expansible curbside). A door to the theater area, located on the curbside on the expansible wall, shall be furnished. The theater area side door shall be as close to the front corner of the curbside expansible section as construction permits. The door opening shall be not less than 27 inches in width and 76 inches in height. The door shall open toward the front. The door construction shall conform to 3.4.3.1.1.

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3.4.3.3 Theater area rear doors (end walls of expansible sides). One rear door shall be provided at the rear of the curbside and one at the rear of the streetside expansible section end walls. The door openings shall be not less than 27 inches in width and not less than 76 inches in height. The doors shall open toward the nonexpansible body sides. The door construction shall conform to 3.4.3.1.1.

* 3.4.3.4 Projection booth rear door. A rear door shall be provided, centered at the rear of the semitrailer, for entry into the projection booth. The door opening shall be 36 inches plus 1, minus 0 inches in width and not less than 78 inches in height. The door construction shall conform to 3.4.3.1.1.

* 3.4.3.5 Double front doors. Double doors shall be provided in the front of the semitrailer. The front door opening shall be of sufficient width for loading and unloading and for access to the air conditioner(s). The door opening shall be not less than 80 inches in height or 4 inches higher than the air conditioner(s), whichever is greater. The door opening shall be adequate to permit removal of the air conditioner(s) for servicing and then reinstallation. The doors shall be not less than the full width of the units and shall be provided with hand operated latches and support rods or other suitable mechanism to hold the doors open when the unit is in service.

* 3.4.3.5.1 Double front door construction. The double front doors shall be side hung on box posts with truck type heavy duty strap type hinges with a grease fitting mounted in the center of each hinge. Means shall be provided to open and retain the doors in a 180-degree full open position. The doors shall be insulated. The doors shall be weatherproof and watertight. The doors shall be provided with three-point, cam type locking devices with latch and key-operated locks. Lock cylinders shall be provided with covers to protect the cylinders from road splash. The latches shall be recessed to prevent intervehicular electric cable and brake air hose interference. Tapered door sills shall be furnished. The two front doors shall contain grilles and dampers to match with the air conditioner opening(s). A removable panel shall be provided to cover these openings during vehicle travel. The air conditioner(s) shall be incapable of being operated until the covers are removed or the double doors are opened. Lock-out switches shall be provided at each of the outdoor coil air intake and discharge doors to prevent air conditioner operation when either door is closed or when the panels are in place. A warning light and notice shall be located beside the air conditioner control, notifying the operator that a panel may cover the condenser. The notice shall be on a plate conforming to 3.1.1.2.2.

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* 3.4.3.6 Service area to theater area partition door. A door shall be provided in the partition between the service area and the theater area. The door shall swing open into the theater area. The door need be operational only when the side sections are expanded. Door opening size shall be not less than 30 by 78 inches.

* 3.4.3.7 Door frames and sills. All door frames and sills shall be constructed of not less than 11 gage (0.1196 inch) material. The door frames shall have at least a 10-degree slope to accommodate uniform clearance to open and close and to prevent entrance of weather, sound and dust.

3.4.3.8 Door gaskets. All door openings shall be provided with light-tight, sealing gaskets conforming to 3.4.1.16. Gaskets shall be fitted around the door openings and, for the front, between the front doors. Gaskets shall have required flexural and compressibility characteristics. When doors are closed, light, wind and rain shall not penetrate.

* 3.4.3.9 Exterior stairways. Five collapsible exterior stairways shall be furnished for entrance to and exit from the service area, the theater area and the projection booth. Each stairway shall be capable of being located perpendicular to its exterior entrance door opening. The stairways (except for the handrails) shall not interfere with opening and closing of the doors. Stairways and handrails shall be capable of being manually handled, fitted in place, removed and stowed in the stowage compartment by one person.

* 3.4.3.9.1 Stairway construction. Stairways shall be of the flat-step type, constructed of aluminum. Means shall be provided in the stairway side members to retain the handrails. The stairway side members shall provide a slope from vertical of not less than 50 degrees. The stairway tread depth shall be a minimum of 9-1/2 inches. The stairway riser height shall be not more than 8 inches. Width between installed handrails shall be a minimum of 36 inches plus 1 inch, minus 0 inches. The standing surface of each stairway shall be of aluminum plate, covered with special abrasive type paint or tape (ship's grip).

* 3.4.3.9.2 Handrails. Removable handrails shall be provided for both sides of each stairway. The height of the handrails shall be 30 to 34 inches, measured from the standing surface of the stairway to the top surface of the handrail. Handrails shall be of aluminum tubing, 1-1/2 inch in outside diameter, and with a wall thickness to withstand the loads specified herein.

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* 3.4.3.9.3 Stairway support. Means shall be provided to support each stairway, at the proper height. Necessary cross-supports shall be provided. Supports shall be provided with shoes providing proper ground contact surface area. A means to level and adjust the stairway assemblies shall be provided.

* 3.4.3.9.4 Imposed loads. Each step of each stairway, each handrail, and the entire assembly shall be capable of accepting the imposed loads specified herein. Each step of each stairway shall be capable of accepting 350 pounds per square foot. Loads shall be imposed at the midpoint of any supporting members of each step. Maximum vertical deflection shall be not more than 0.26 inch midway between supports. The stairs, supports and brackets shall be capable of supporting a total static load of not less than 6,000 pounds. The handrails shall be capable of supporting, vertically and longitudinally, 3 personnel of 200 pounds each leaning on the handrails, without any deformation of the handrail, supports or fastenings.

* 3.4.3.9.5 Fasteners. Fasteners shall be provided to secure the handrails to the side members of the stairways and to the semitrailer entrances. Fasteners shall be corrosion-resistant and shall be capable of securing all members in not more than 15 minutes by one man per stairway using common hand tools.

3.4.3.9.6 Safety. All sharp edges shall be rounded to remove any sharp edges. Rolled-over burrs from cutting operations are not acceptable. Entrance to and exit from the semitrailer shall not present any tripping hazards.

* 3.4.4 Stowage compartment. A stowage compartment shall be furnished under the body. The stowage compartment shall be bolted to frame structural members. The compartment shall be suitable for use in stowing and of adequate size to stow the truck tractor spare tire, the semitrailer spare tire, spare tire tools, stairs, handrails, fasteners, extension ladder, chock blocks, landing gear blocks, electrical cable assemblies, cable reels, and all tools required to operate the semitrailer and the engine-generator that is mounted behind the cab of the towing vehicle. Means shall be provided to securely retain each of the specified items in the stowage compartment. The stowage compartment shall extend from the curbside to the streetside and shall provide not less than 14 inches of road clearance. The total length of the compartment shall be limited only by the clearance required for the rear tires and by the front ground clearance (see table I), and shall be the maximum length practicable. The stowage compartment shall be waterproof.

* 3.4.4.1 Sections in the stowage compartment. There shall be four or five sections in the stowage compartment, extending the full length

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from the rear of the landing gear to the front of the tires. Each section shall have individual doors on both sides and shall be equipped with recessed, round, paddle type, stainless steel latches and key operated locks. Latches and locks selected by the contractor shall be subject to pre-approval/rejection by the Government. Each section shall be not less than 47 inches in width, and not less than 18 inches in height, except the width of the forwardmost section may be reduced as necessary for required ground clearance (see table I).

* 3.4.4.2 Stowage compartment construction. The stowage compartment shall be constructed of welded construction. Corners shall be reinforced with structural members. Doors shall be double paneled, reinforced. The stowage compartment shall have a load-carrying capacity of not less than 50 pounds per square foot and greater as required to accomodate the stowed equipment specified herein. Hinges shall be continuous and shall be provided with stainless steel sockets. Stowage compartment streetside and curbside doors shall be hinged at the top. Means shall be provided to positively retain the doors vertically in the open position whether the expansible sides are retracted or expanded. Doors shall be waterproof.

* 3.4.4.3 Stowage compartment lighting. Each section of the stowage compartment shall be provided with a 12-volt and a 120-volt lighting fixture. Lamps shall be provided with guards. Lighting fixtures shall be protected, waterproof and readily accessible for servicing. Lamp bulbs shall be capable of being replaced without removing the fixtures. The front curbside and the front streetside compartment sections shall be provided with recessed or flush mounted, three-way, 12-volt switches controlling all 12-volt compartment lights. Switches shall be waterproof. The 12-volt system shall conform to 3.4.8.1. The 120-volt system shall conform to 3.6.5.4.

3.4.5 Extension ladder. An aluminum extension ladder shall be provided for cleaning and servicing the expansible roof sections. The ladder shall be of sufficient height for personnel to climb up to the roof.

* 3.4.6 Semitrailer running gear.

* 3.4.6.1 Suspension system. The semitrailer shall be furnished with a tandem axle, air ride suspension system, with each component having a rated capacity at least equal to the load imposed, measured at the ground, with the vehicle loaded as specified in 3.2.3 through 3.2.5.1. When the semitrailer is operating over rough terrain, on slopes and on the highway, the underside of the body shall not touch the tires. In the event of loss of air pressure by the air suspension system, the tires shall not contact any portion of the underside of the semitrailer body. Rubber mounted components shall be provided. Torsion

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bars to control sway shall be provided. Hydraulic shock absorbers shall be provided. The suspension system shall include an air control system with manually operated spring control valves on the curbside and streetside of the bogie to maintain semitrailer height and to provide leveling of the semitrailer and shall not allow the tires to depart from the ground when the semitrailer is supported by the four landing gear assemblies. An air filter shall be provided at each air spring control valve. The air supply system shall provide equal amounts of air to each air bag under travel conditions. The system shall conform to 3.1.1.16. In addition to the air brake reservoir(s), a reservoir for the air suspension system and the air operated upper fifth wheel kingpin assembly shall be furnished. The reservoir shall have a capacity of not less than 3,000 cubic inches. The reservoir shall include at least the following components:

(a) Pressure protection valve
(b) Check valve
(c) Automatic moisture ejector valve and drain cock. The drain cock location shall be marked as to its lateral and longitudinal location on the exterior of the semitrailer
(d) Air filter, in addition to the air filters at each air spring control valve
(e) Air supply hose complying with SAE J1402
(f) Air system schematic on a plate conforming to 3.1.1.2.2
(g) Grease fittings on all pivot or bearing areas.

* 3.4.6.2 Air mounted fifth wheel kingpin assembly. An air mounted fifth wheel kingpin assembly shall be installed. The upper fifth wheel assembly shall be suitable for coupling to a fore and aft rocking 36-inch diameter fifth wheel on a towing vehicle. The upper fifth wheel assembly shall be in accordance with Federal Motor Carrier Safety Regulations 393.70(b). A 2-inch, SAE J700, alloy steel kingpin, hardened and ground, shall be provided. The forward end of the upper fifth wheel assembly shall have 30-degree chamfer, with a 4-inch turned-up lip for ramping purposes when coupling to a towing vehicle. The 4-inch turned-up lip shall be secured only to the width of the air mounted upper fifth wheel assembly pick up plate. The upper fifth wheel assembly shall have a natural frequency of 99 cycles per minute at 32,000 pounds static load. The fifth wheel jaws of a truck tractor fifth wheel shall mate with the kingpin without any evidence of galling the fifth wheel jaws or the kingpin recessed 2.000 inch diameter. The assembly and its components shall be as recommended by the manufacturer of the air mounted fifth wheel kingpin assembly. The system shall conform to 3.1.1.15. The kingpin shall be located as specified in table I. The upper fifth wheel assembly shall include at least the following components:

(a) Leveling valve with air filter
(b) Shock absorbers (2 each)

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(c) Grease fittings for the rubber mounted bearings at pivot areas
(d) Air bags (4 each)
(e) Air controls
(f) Pressure protection valve, with air filter
(g) Air supply tubing complying with SAE J844
(h) Air system schematic on a plate conforming to 3.1.1.2.2.

* 3.4.6.3 Axles. Tandem rear axles shall be furnished. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded as specified in 3.2.3 through 3.2.5.1. Wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hubs shall be fitted with leakproof seals. The hub caps shall have a window for visual determination of the oil level. Provisions for venting and replenishing the oil supply shall be incorporated. Axle alinement shall be in accordance with SAE J875.

* 3.4.6.4 Brakes. Full air, internal type brakes shall be furnished on all the wheels of the semitrailer. The brake system shall have a rated torque capacity at least equal to the maximum torque imposed during braking, measured at the ground, with the semitrailer loaded with rated payload. The brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.50, and Federal Motor Vehicle Safety Standard No. 121. All components of the brake system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake system components. No part of the braking system shall extend below the bottom of the wheel rims. Whenever possible, air chambers and slack adjusters shall be mounted above the bottom edge of the axle carrier. The location of air fittings shall be in accordance with SAE J702. Air fittings shall be mounted in recessed locations. Dummy couplings with security chains shall be furnished. The gladhand seals shall be treated with silicone conforming to MIL-S-8660 and QPL 8660. The gladhands shall be in accordance with SAE J318. See 3.1.1.15.

3.4.6.4.1 Brake components. The brake system shall be complete and shall include at least the following:

(a) Air reservoir, end drain design, having a capacity of not less than 12 times the brake chamber capacity at maximum stroke;
(b) Automatic reservoir drain valve or drain cock. Drain cock shall be marked as to lateral and longitudinal location on the exterior of the vehicle and shall be accessible from the side of the vehicle;
* (c) Automatic slack adjusters;
(d) Air actuators for service, parking and emergency braking;

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(e) Air brake tubing, air suspension tubing and air operated fifth wheel tubing to comply with SAE J844;
(f) Identification tags, emergency and service;
(g) Relay emergency valve, 4-way;
(h) Single safety valve to prevent air pressure back-up in the reservoir.

3.4.6.4.2 Parking brakes. The parking brakes shall be of mechanical, spring actuated type, acting on the service brakes of all wheels, and shall be designed to operate in conjunction with the service brake system. The brake actuating mechanism shall apply the parking brakes when the air line pressure drops below the normal operating pressure. Provisions shall be included to permit release of the parking brakes when compressed air is not available to release the brakes. Parking brakes shall comply with Federal Motor Carrier Safety Regulations 393.41 and Federal Motor Vehicle Safety Standard No. 121. See 3.1.1.15.

* 3.4.6.5 Wheels, rims and tires. The vehicle shall be equipped with dual wheels on each axle. Rim and tire ratings shall conform to Tire and Rim Association recommendations for the type and size tires furnished. Tire size, load range (ply rating), tread design and manufacture shall be the same for all wheels and tires on the vehicle. Rim and wheel load ratings shall meet or exceed the loads imposed with the vehicle loaded as specified in 3.2.3 through 3.2.5.1. Polished aluminum disc type wheels shall be furnished. Minimum dual spacings shall be as specified by the Tire and Rim Association for the size tire required and shall conform to the Federal Motor Vehicle Safety Standards No. 119 and No. 120.

* 3.4.6.6 Tires. Tires shall be steel belted radial tubeless type with highway tread. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded as specified in 3.2.3 through 3.2.5.1 and at the speeds specified in 3.3.1. Tires shall conform to Tire and Rim Association recommendations, or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association. Tires shall conform to Federal Motor Vehicle Safety Standard No. 119.

3.4.6.7 Spare tire assembly. A spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, load range, and tread design as the tires installed on the vehicle.

3.4.6.8 Wheel and tire balancing. Hubs, drums, tire and wheel assemblies shall be balanced to prevent abnormal tire wear, wheel thump and drumming at speeds specified in 3.3.1.

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* 3.4.6.9 Hubodometer. A hubodometer shall be installed at one of the curbside wheels.

3.4.6.10 Tools. The semitrailer shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of a wheel and tire assembly. Tools shall be stowed in the stowage compartment. Wheel chocks for the front and rear of the rear tandem tires shall be furnished and stowed in the stowage compartment.

* 3.4.7 Landing gear assemblies. The vehicle shall have four vertical lift, telescopic, nonrotating landing legs with two speed gears, and a hand crank for each landing leg. The landing legs shall be equipped with self-leveling skid pads. A holder shall be provided for each hand crank. The landing gear assemblies and components, accessories, and attachments shall withstand, without deformation, the combined static and dynamic forces due to the portion of the gross weight sustained, and the additional forces resulting from impact during coupling and uncoupling operations. When placed in travel position, the landing gear assemblies shall remain positively locked. The landing gear assemblies shall be protected to prevent the entrance of foreign matter which would impair its functioning or mechanical efficiency. Lubrication fittings shall be furnished. The front landing gear assemblies shall have sufficient vertical travel to vary the height of the upper fifth wheel plate, measured at the vertical centerline of the kinpgin, from at least 47 inches to not less than 55 inches above ground level, without blocks. Two blocks, each 3 inches in height, shall be furnished for the two front landing gear assemblies for additional variation in the vertical travel. The two front landing gears shall be located so as to conform to the landing wheel clearance specified in table I. Two landing gears shall be located rearward of the tandem assembly, one on the curbside and one on the streetside. The four landing gear assemblies shall be suitable for use in side-to-side and fore and aft leveling the semitrailer, with the air suspension inflated and the tires not departing from the ground. Landing gear shall be adequate to enable the semitrailer to be leveled to the degree required by the performance requirements of 3.3.10. Clearance under the rear landing gear shall be such that the landing gear does not reduce the normal rear departure angle of the semitrailer.

* 3.4.7.1 Landing gear assemblies, front, air operated. The two front landing gear assemblies shall be capable of being air-operated in addition to the manual operation specified in 3.4.7. The two front

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landing gears shall be controlled by a twist-and-push, on-and-off button. The control shall reduce false starts. The system shall have a manual override cranking system for use when a truck tractor air supply is not readily available. The air operated system shall automatically shut off or shall ratchet when the landing gear is fully retracted or when the landing legs are firmly on the ground. The system shall provide two speeds. An individual air reservoir with condensation drain shall be provided. The air operated landing gear mechanism shall not extend beyond the width of the semitrailer nor shall it be subject to damage by coupling and uncoupling operations. See 3.1.1.15.

* 3.4.7.2 Spirit levels. Four spirit levels shall be furnished. The levels shall be mounted in protected but visible locations. The levels shall indicate the degree of body level when the landing gear assemblies are being employed as leveling jacks. The levels shall be resistant to inclement weather and shall be corrosion-resistant. The levels shall be mounted, one on the front and one on the rear of the body, and one on the streetside and one on the curbside, equidistant between the front and rear landing legs. The levels shall be recessed. The glass shall be unbreakable. Level accuracy shall be adequate to enable the semitrailer level to be measured to the degree of accuracy required by the performance requirements of 3.3.10.

* 3.4.8 Electrical system, 12-volt. The exterior electrical system that operates the vehicle running lights shall be 12-volt dc and shall conform to Federal Motor Vehicle Safety Standard No. 108. The brake lights shall override the four-way emergency flasher.

* 3.4.8.1 Wiring and conduit, 12-volt. Wiring shall be in accordance with the applicable portion of SAE J1292, and in accordance with Federal Motor Carrier Safety Regulations 393.27, 393.28, 393.29, 393.32 and 393.33. All conductors shall be copper. All wiring exposed to possible physical damage shall be protected by use of heavy wall thermoplastic insulated conduit or aluminum conduit. Wiring within the body area that is not readily accessible for service shall be in conduit. Conduit shall be provided with vents. The number of conductors permitted for each conduit shall be determined by the overall diameter of the cable or conductor with the waterproof cover installed. All fasteners to the conduit and conduit fittings shall be of a material that shall not exhibit corrosion and shall be chemically compatible with the surfaces that contact the fitting and conduit and with the surface to which they are attached or secured. All exterior lighting shall be wired within the body to exterior lighting receptacles. Wiring shall be accessible for replacement without removing any component of the vehicle. Flexible metal conduit is not acceptable. Wiring shall be capable of being replaced and serviced through the conduit and junction boxes. Conduit shall be adequately and properly secured to prevent possible physical damage, by the employment of chafeproof clamps and

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grommets. Grommets or electrical bulkhead fittings shall be employed for any conduit passing through any structural member. All 12-volt wiring shall be color-coded as specified in SAE J560.

3.4.8.2 Exterior lighting, 12-volt. All lights and reflectors shall be mounted in recessed locations or in guarded locations to provide protection from operational hazards. Lights and reflectors shall not be mounted on rub rails (unless recessed and fully protected) or on vehicle bumpers.

* 3.4.8.3 Receptacle for 12-volt system. The vehicle shall be equipped with a 12-volt lighting receptacle mounted in a recessed location at the front of the vehicle. The receptacle shall be located in accordance with SAE J702. The receptacle shall be in accordance with SAE J560, with spring-loaded cover assembly. The 7-contact socket shall be wired in accordance with SAE J560 to supply 12 volts directly to the semitrailer circuits.

* 3.4.9 Expansible side sections. Expansible side sections shall be provided on the streetside and on the curbside of the semitrailer. Entrance to and exit from the expansible sides shall be by the exterior curbside door specified in 3.4.3.2, by the two exterior rear doors specified in 3.4.3.3 and by a partition door from the service area to the theater area as specified in 3.4.3.6. Each expansible side shall consist of front, rear and side walls, floor and roof, all integrated with the gear racks (see 3.4.9.5) to form a rigid, 5-sided, self-supporting box structure, meeting the requirements of 3.3.9. Each expansible side section shall be rigidly constructed into a single assembly with appurtenances for expanding the two expansible sides simultaneously as a single unit. The design and construction of the expansible side sections shall conform to 3.4.1.10 through 3.4.2.1. The sides, end and roof of the expansible sections shall be insulated to the same thickness as the nonexpansible portions of the semitrailer with insulation conforming to 3.4.1.13. The floor of the expansible sections shall conform to 3.5.3. In addition, the aluminum structural subframe required by 3.5.3 shall provide an insulation cavity not less than 2 inches high, immediately below the plywood. The cavity shall be fully filled with insulation conforming to 3.4.1.13, protected by a floor pan conforming to 3.4.1.2. The bottom, exterior surface of the expansible sections shall be equal in design, construction, workmanship and appearance to the exterior body panels of the semitrailer (see 3.4.1.10). As an alternate to the folding floor in the nonexpansible area, as required by 3.4.1.4, a fixed floor in the nonexpansible area and a folding floor in the expansible side sections may be provided. Expansible side section folding floors shall be hinged in the area of the normal side walls of the semitrailer and not in the expansible sections. Otherwise the floor, including subframe and insulation, shall be as required above and as required by 3.4.1.4 and 3.5.3.

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* 3.4.9.1 Actuation. The expansion and retraction of the expansible sides shall be actuated by two systems, one by manual cranking and the other by an electric motor which shall operate the expansible gear mechanisms. A safety system shall be installed to prevent operation of the expansible van electric motor with any section of the folding floor in the down position. Means shall be provided to prevent over-expanding the expansible side sections, manually and with the electric motor.

3.4.9.2 Power transmission (see figure 12). Each expansible side section shall be actuated by four gear racks secured to each expansible side section. The eight gear racks (four each side) shall be actuated by drive gears that engage the eight racks. One drive gear shall be mounted on each end of four vertical drive shafts (two each side). A retaining bearing block shall be secured at the top of the vertical shafts as a stop. The vertical drive shaft shall be actuated by speed reducers. The input shafts of the speed reducers shall each be provided with a flexible coupling. The speed reducers shall be driven by two horizontally mounted miter boxes. The forward miter box shall have one input shaft, actuated by the rear miter box output shaft, and two output shafts that actuate the forward speed reducers. The rear miter box shall have one input shaft and three output shafts. Two output shafts shall actuate the rear speed reducers and one output shaft shall actuate the drive shaft to the front miter box. The main drive shaft from the rear to the front miter box shall be located under the nonexpansible body.

3.4.9.3 Shafting. All shafting shall have sufficient strength to prevent torsional problems. The shafting shall be suspended by self-alining pillow blocks with waterproof seals and bearings. Means shall be provided for lubrication. Shafting shall be hardened and ground stock to prevent galling in the pillow block bearings. Boots shall be provided to prevent road splash on the main drive shaft. A lubrication manifold to each pillow block shall be provided in the stowage compartment. Means shall be provided to adjust the retaining bearing block located on the shaft in the event the expansible sides require recentering or timing.

* 3.4.9.4 Speed reducers. Speed reducers shall be the correct model for either vertical or horizontal application. Speed reducers shall be mounted on steel plates and shall be bolted to the plates.

* 3.4.9.5 Miter boxes. Miter boxes shall have a ratio of 1:1 and shall withstand the loads imposed by the input forces developed during actuation. The rear miter box shall be accessible at the rear of the semitrailer.

* 3.4.9.6 Gear racks. Gear racks shall be provided to actuate the expansible sides. The gear racks shall mate with the drive gear teeth

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without evidence of binding or deformation. The expansible sides shall roll on stainless steel or oil treated bronze rollers, mounted on ball or roller bearings. The roller cavity shall be coated with teflon to prevent galling of the metal cavity and to prevent water freezing.

3.4.9.7 Manual drive. A drive shaft with a 12-inch hand crank shall be provided to actuate the rear miter box which in turn shall actuate all other miter boxes and speed reducers. Maximum effort to expand or retract the side sections shall not exceed 40 pounds of effort on the crank handle. The crank shall be capable of rotating 360 degrees continuously without being removed from the drive shaft. The crank shall be capable of ratcheting clockwise and counterclockwise and of rotating without the use of the ratchet mechanism.

3.4.9.8 Electric drive. In addition to the manual drive, the expansible side sections shall be capable of expanding and retracting electrically in accordance with 3.6.8.

3.4.9.9 Alinement. The actuating mechanism shall have a means provided for alinement. A movable stop shall be provided. All gearing and shafting shall be tangent to each other. No deformation shall develop.

* 3.4.9.10 Power transmission maintenance. All parts of the power transmission system shall be protected from road splash and road hazards, shall be shielded within the stowage compartment against interference with operations by stowed equipment, and shall be accessible for lubrication (see 3.3.11) and maintenance, including replacement. Means shall be provided for access to all power transmission components, parts and assemblies (see 3.4.9.2 through 3.4.9.9) without removing the expansible side sections for the servicing and replacement of all parts, assemblies and components.

3.5 Interior. The semitrailer shall be divided by partitions into three areas: the service (front) area; the theater area (consisting of the area when the sides are expanded); and the projection booth at the rear of the semitrailer (see figure 7).

3.5.1 Lateral partitions. Lateral partitions shall be provided as depicted in figures 7, 8 and 10. The service area partition shall be forward of the theater area and shall provide a door for entrance to the theater area when the theater side sections are expanded. The projection booth partition shall be rearward of the theater area (see figure 8), and similar in construction to that depicted in figures 9 and 10. The projection room partition shall be insulated, acoustically treated and permanently installed between the theater area and the projection booth. Three 1/4-inch thick glass windows, 20-inches by 24-inches of tempered water pane optical safety glass, shall be provided

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(see figure 10). The glass shall be installed so that the glass will withstand the shock and vibration of travel mode as specified in 3.3 through 3.3.8.

* 3.5.2 Ceiling. The ceiling for the three areas shall be covered with perforated, aluminum, acoustic sheets. A wireway shall be installed in the ceiling (see figure 6 and 3.5.5.6). The ceiling sheets in the theater area shall be supported adequately to have a hanging capacity of not less than 25 pounds on any one square foot of area and a total of not less than 1,000 pounds for the entire theater ceiling area, to permit displays to be suspended from the ceiling.

* 3.5.3 Flooring. Flooring shall consist of marine grade A/C plywood, sanded on the top side. Flooring shall be at least 3/4 inch thick, backed by a suitable aluminum structural subframe. All seams shall be sealed. Provisions shall be made for installation of 1/2-inch thick carpet in the entire theater floor area by the receiving activity. When the folding flooring is unfolded, it shall form one plane with the fixed floor (see above, 3.4.1.4 and 3.4.9), without evidence of any protrusion.

3.5.4 Walls. The walls of the three areas shall be 3/8-inch minimum thickness, marine grade, A/C plywood, (see 3.1.1.3) sanded on the exposed side. Joints shall be sealed. The walls of the theater area will be covered with carpet 3/8 inch to 1/2 inch thick by the receiving activity. Electrical receptacles and cover plates shall be positioned so that the cover plates will allow for the installation of the 1/2-inch thick carpet around the receptacles.

3.5.5 Accessories.

3.5.5.1 Threshold plates. Threshold plates shall be provided at each door opening. The plates shall be of aluminum or stainless steel.

3.5.5.2 Door kick plates. Each door casing shall be provided with stainless steel or aluminum door kick plates.

* 3.5.5.3 Door trim. Each door shall be provided with door trim across the top of the door and on the vertical sides. In addition, a polished aluminum plates shall be furnished at the door handles. Door handles shall be recessed.

* 3.5.5.4 Locks and keys. Locks shall be provided for all body doors and for all stowage compartment sections. Locks shall be heavy duty. All body door locks shall be operable by the same key. All stowage compartment section door locks shall be operable by the same key. Three rings of keys shall be provided. Latches and locks shall be designed and installed so that they cannot be removed with a screwdriver or a wrench when the door is closed and locked.

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3.5.5.5 Removable floor panels. Two removable floor panels for use near the front and rear walls of the expansible sections shall be provided. Each floor panel shall have two recessed rings to enable personnel to lift the floor panel from its installed location prior to retracting the expanded sections. When in use, the floor panels shall be in the same plane as the interior floor. An aluminum or stainless steel moulding shall be provided around the perimeter of both floor panels. A stowage rack shall be provided for both floor panels.

* 3.5.5.6 Wireway. Wireway compartments shall be provided in the ceiling for five 1-inch conduits (see figure 8). One conduit shall extend from 6 inches into the projection booth to 3 feet rearward of the service area partition. Four 1-inch conduits shall extend from 6 inches into the projection booth to 6 inches into the service area. The wireway shall be at least 2-3/4 inches deep and 12 inches wide. Panels shall be provided in the ceiling, secured with 1/2-turn, aircraft, corrosion-resistant fasteners. The panels shall be framed with a "U" shaped stainless steel or aluminum moulding around the perimeter and shall be secured to resist vibration. Panels shall be provided wherever the wireway projects into the projection booth, the service area and the theater area. Each panel shall be provided with a fingerhold that personnel may use to pull the panel out and use when releasing or securing with the 1/2-turn fasteners.

* 3.6 Electrical systems, 120/240 volt. A 240-volt, 60 Hz, 5-wire, three phase and a 120-volt, 60 Hz, 3-wire, single phase electrical system of sufficient amperage capacity to meet all the electrical loads imposed shall be provided. The system supply will normally be a diesel engine driven, 40 kW generator mounted on the towing vehicle. All wiring and cable conductors shall be copper. The neutral and equipment ground shall be completely separate and not tied together anywhere. All electrical equipment housings, frames, casings, etc., shall be grounded with a green wire with continuity to the equipment ground at the power source. All wiring shall be in properly sized metal wireways, raceways or conduit. Exterior wiring shall be in conduit. The number of conductors permitted for each conduit shall be determined by the overall diameter of the cable or conductor. All fasteners to the conduit and conduit fittings shall be of a material that shall not exhibit corrosion and shall be chemically compatible with the surfaces that contact the fitting and conduit and with the surface to which they are attached or secured. See 3.2.8.

* 3.6.1 Cable, 240-volt. A minimum of 100 feet of 240-volt cable shall be provided. Cabling shall be heavy duty, low temperature, oil- and heat-resistant, extra flexible. Cables shall be capable of being handled by personnel in an ambient temperature from minus 20°F to plus 160°F without evidence of deformation or loss of dielectric properties, and shall be capable of winding and unwinding on the cable reel at minus

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20°F without cracking. Conductors of the cable shall be stranded copper and shall meet or exceed the electrical load imposed. Color of individual conductor insulation shall be black, white, red, green and orange. Black, red and orange shall be phase conductors, white shall be neutral and green shall be equipment ground. Conductor size shall be adequate as specified herein. Manufacturer's certification for the cables shall be available. Dealer's and distributor's certification shall not be acceptable.

* 3.6.1.1 Cable terminations. The 240-volt cable shall be furnished with a 120/240-volt termination on each end. Terminations shall meet amperage and voltage requirements. One end of the cable shall be terminated with a plug having live front contact pins and the other end terminated with a connector having dead front contact tubes. In use, the plug shall mate with the power source receptacle on the engine generator and the connector shall mate with the inlet on the semitrailer. Cable terminations shall be watertight when in use and when stowed.

* 3.6.2 Cable, 120-volt. In addition to the 240-volt cable, a minimum of 100 feet of 120-volt cable shall be provided. Cable shall be stranded copper wire, heavy duty, low temperature, oil- and heat-resistant, and extra flexible. The cable shall be capable of being handled by personnel in an ambient temperature from minus 20°F to plus 160°F without evidence of deformation or loss of dielectric properties, and shall be capable of winding and unwinding onto the cable reel at minus 20°F without deformation. The cable shall be rated at not less than 60 amperes.

3.6.3 Power entrance box, 240-volt. A recessed 240-volt power entrance box shall be installed in the semitrailer at the front curbside. The power entrance box shall be furnished with a hinged, gasketed cover with a means to hold it open and with a fastener to retain it closed. The entrance box shall be marked with a plate in accordance with 3.1.1.13.

* 3.6.3.1 Power inlet, 240-volt. The power inlets shall have live front contact pins and shall be rated at 240 volts. Power inlets will mate with the dead front connections of the main power cable as specified in 3.6.1.

3.6.3.2 Power entrance box, 120-volt. A recessed 120-volt power entrance box shall be installed in the semitrailer, curbside front. The power entrance box shall be furnished with a hinged gasketed cover with a means to hold it open and a fastener to retain it closed. The entrance box shall be marked with a plate in accordance with 3.1.1.13.

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3.6.3.3 Power inlets, 120-volt. The 120-volt power inlet shall have live front contact pins. Power inlets shall mate with the dead front connections of the 120-volt cable specified in 3.6.2.

* 3.6.4 Main power distribution panel. The main power distribution panel shall be rated at 240 volts and 120 volts. The panel(s) shall provide not less than 20 circuits of 120 volts, 60 Hz, of 20 amperes each for general purposes (see 3.6.5); 3 circuits of 120 volts, 30 amperes for the three voltage regulators in the projection booth; 2 circuits of 120 volts, 30 amperes for receptacles in the projection booth; a 120-volt, 20-ampere circuit for the stowage compartment lights and receptacles; and a 120-volt, 30-ampere circuit for the empty wireway in the ceiling. Two individual or one combined 220-volt circuit(s) for the air conditioning and heating units shall be provided. An individual 220-volt circuit for the electric motor to power the expansible side sections shall be provided. All circuits shall be provided with thermal-magnetic, bolt-in circuit breakers, each with the required ampere rating for the panel board for overcurrent protection. The panel shall be wired to provide 120-volt power and 240-volt power, 60 Hz ac. Circuit breakers shall be of the ground fault interrupter type. The green wire and the white wire shall be separate and shall be identified.

* 3.6.5 Circuits, 120-volt. The vehicle shall be provided with 120-volt circuits, duplex receptacles, lighting and switches. Switches and receptacles shall be recess-mounted (except see 3.5.4), matching brown in color and shall be provided with matching polished aluminum cover plates. Switches and receptacles shall have the proper voltage and current ratings. The receptacles and lighting shall be grouped on individual circuits as follows:

(a) Service area: Three 20-ampere receptacles, one on the curbside forward of the entrance door, one on the streetside directly opposite, and one approximately in the center of the air conditioner partition, facing into the service area. Receptacles shall be located 18 inches, plus or minus 1/2 inch, above the floor and shall be recessed.
(b) Projection booth: Two 20-ampere receptacles, one on the streetside and one on the curbside of the projection booth, 18 inches, plus or minus 1/2 inch, above the floor level (see figure 10). Two 30-ampere receptacles shall be similarly located. The 30-ampere receptacles shall be identified "30 AMPERE" with a plate conforming to 3.1.1.2.2.
(c) Theater area: Both side walls of the expansible sides shall have six 20-ampere duplex receptacles. Three receptacles on a

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single top row on each expansible side shall be on one circuit. Three receptacles on a single bottom row shall be on one circuit. Location shall be as noted on figure 11. The receptacles shall be so recessed that, when the receptacle covers are installed, they will be flush when 1/2 inch of carpet is installed on the wall.

(d) Lighting: Each row of lighting shall be on a separate circuit.

(1) One row of incandescent lighting fixtures shall be securely attached to both expansible section ceilings not less than 6 inches from the side walls and shall be secured to prevent shifting during transit. Lighting fixures shall have acrylic prismatic diffusers hinged at one side of each row with a stainless or aluminum continuous type hinge and shall have positive securing catches on the side opposite the hinge. Covers shall be hinged so that the cover drops to the side wall when opened. Lamps shall be provided.

(2) Fluorescent lamps for the theater area shall be provided. The lamps shall be capable of operating at 0°F ambient temperature without loss of illumination. The lamps shall be of standard design, nominally 48 inches in length. The lamps shall be 40-watt, type T-12, bipin, cool white, rapid-start lamps. The lamp fixtures shall be of the commercial type, recess-mounted into the ceiling (see figure 6). The fixtures shall have an acrylic prismatic diffuser hinged at one side with a continuous type hinge and shall have positive securing catches on the other side. At least two captive screws or similar devices shall be employed on the side opposite the hinge to assure that the cover does not drop during transit. The fixtures shall extend the entire length from the service area partition to the projection booth partition. Two parallel rows of fixtures shall be provided as depicted in figure 6.

(3) Two continuous rows of fluorescent light fixtures (see (2) above) shall be provided in the service area and in the projection booth.

(e) Wiring to the expansible sections shall expand and retract with the expansible section.

* 3.6.5.1 Light switches. Illuminated light switches shall be provided for all lighting as follows:

(a) Center theater lights - illuminated three-way switches, one adjacent to the rear projection booth door and one adjacent to the service area-to-theater area partition door, on the theater side.

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(b) Theater area side wall lights - illuminated three-way dimmer switches, two adjacent to the rear projection booth door and two adjacent to the service area-to-theater area partition door, on the theater side.
(c) Center service area and projection booth lights - illuminated switches in the service area or projection booth, as applicable.
(d) All six theater light switches shall be identified "CURBSIDE THEATER LIGHTS," "CENTER THEATER LIGHTS," or "STREETSIDE THEATER LIGHTS," as applicable, with plates conforming to 3.1.1.2.2.

* 3.6.5.2 Circuits for projection booth voltage regulators. Three 30-ampere junction boxes for the 30-ampere voltage regulator circuits shall be provided. The circuits shall be separate. The junction boxes shall be located two on the streetside and one on the curbside, forward of the voltage regulators. The receptacles shall conform to Underwriters Laboratories, Incorporated (UL) 498, Figure 65-5. The receptacles shall be identified "30-AMPERE" with a plate conforming to 3.1.1.2.2.

* 3.6.5.2.1 Voltage regulators. Three single phase, 120-volt, 60 Hz voltage regulators shall be located in the projection booth, two on the streetside and one on the curbside, two inches from the front and side walls. The voltage regulators shall be floor mounted and shall be secured with quick-release brackets. The brackets shall include a means to prevent the voltage regulators from vibrating during mobile operations. Each voltage regulator shall have a 30-ampere, 120-volt receptacle and a 20-ampere, 120-volt receptacle. The receptacles shall be capable of accepting L5-20 and L5-30 plugs, in accordance with Figure 64-6 and Figure 65-5 of UL 498. The L5-20 receptacles on the voltage regulators may be used for any 20-ampere appliance. The output power of each voltage regulator shall be 2.0 kVA. Percentage output shall be 117 volts ac plus or minus 7 percent 60 Hz. The input shall be 95 to 135 volts ac. The voltage regulators shall protect equipment from brownouts and voltage variations and shall remove spikes and noise from the power lines. Response time shall be less than one cycle; low output impedence; 94 percent efficient; low coupling capacity 0.001 pF; negligible distortion; 0°C to 40°C ambient temperature range; low audible noise of 40dB maximum on the ANSI "A" scale.

3.6.5.3 Voltage and frequency meters. A voltage meter and a frequency meter shall be installed in the projection booth, on the curbside wall. The voltage meter shall be automatic, having a range of 0-250 volts ac. The frequency meter shall be automatic ranging from 0-100 Hz. Necessary conduit shall be securely fastened to the interior

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of the outside wall in the projection booth. Meters shall be wired in-line between the power outlet of the voltage regulator and the receptacle outlet of the voltage and frequency meters.

* 3.6.5.4 Stowage compartment 120-volt lighting and receptacles. 120-volt lighting shall be provided for each section of the stowage compartment. In addition, an electrical duplex receptacle with a gasketed, spring-loaded cover shall be provided in the front curbside and in the front streetside sections. Receptacles shall be rated for 20 amperes. Fixtures and lamps shall be protected and waterproof. A three-way, waterproof switch shall be located in the front curbside and in the front streetside sections, controlling all 120-volt compartment lights. In addition, each section shall be provided with 12-volt dc lighting (see 3.4.4.3). The 12-volt and the 120-volt wiring shall be in separate conduits.

3.6.6 Cable reels. Cable reels of sufficient size to accommodate the cables specified in 3.6.1 and 3.6.2 shall be provided. The cable reels shall be of the hand-crank type and shall be capable of storing the 100 feet of 240-volt cable and the 100 feet of 120-volt cable. Cable reels shall be located in the stowage compartments.

3.6.7 Expansible sides, electric driven. Expansion and retraction of the expansible sides shall be electrically activated by a 220-volt electric motor, in addition to being expanded and retracted manually with a hand crank. The electric motor shall be in accordance with UL 674. The motor shall be totally enclosed and fan-cooled. Actuation of the electric motor shall be by a starter push-button located in an accessible, weatherproof and protected location. Means shall be provided to prevent over-expansion of the sides.

* 3.6.8 Ground rod. A spiked ground rod and not less than 25 feet of grounding cable shall be provided for grounding the semitrailer when it is connected to a commercial power line. A plate warning that the vehicle must be grounded when so used shall be mounted near the power entrance boxes. The plate shall conform to 3.1.1.2.2.

3.7 Battery powered emergency lighting. Battery powered emergency lighting shall be installed. Not less than one emergency light shall be located in the service area, one in the projection booth and six in the theater area. The lighting fixtures shall be positioned to provide maximum illumination of each area. Emergency lighting shall be powered from a self-contained power pack in each fixture which operates a 40-watt lamp, or from a single power pack, providing power to all eight emergency lights. The emergency lighting system shall be powered from maintenance-free, rechargeable, sealed, nickel cadmium type batteries. Batteries shall be kept automatically charged by a solid state charger.

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The system shall be provided with pilot light(s) and test button(s) located at the battery(ies) for periodic system checks. Fixtures shall be wired as switched units. The battery charger shall be connected to an unswitched 120-volt circuit and shall conform to 3.6. The emergency lights shall become operational automatically each time there is loss of power.

3.7.1 Battery charger(s). The battery charger(s) shall be 120-volt, completely automatic and shall be capable of restoring batteries to capacity within six hours after restoration of the utility power. The battery charger system shall conform to 3.6.

3.8 Interior 12-volt lighting system. Four 12-volt dc domelights shall be provided in the ceiling. Two domelights shall be located centrally in the theater area, one in the service area and one in the projection booth. The lamps shall be wired to operate from the towing vehicle 12-volt electrical system. A switch shall be provided on the body interior wall at the rear projection booth door. The switch shall be flush or surface mounted, brown in color, and shall have a polished aluminum coverplate. The switch shall be labeled "12-VOLT INTERIOR LIGHTS" with a plate conforming to 3.1.1.2.2. The system shall conform to 3.4.8.1.

* 3.9 Air conditioning and heating systems. The air conditioning and heating systems shall consist of 220-volt, vertical type units mounted at the front of the semitrailer, immediately rearward of the front double service doors and forward of an insulated partition to be installed in the service area (see figures 7 and 8). The units shall be designed and constructed for use in mobile vehicle operations as specified herein (see 3.3.8). The total air conditioning and heating systems shall be designed to control space temperature at 73.4°F (23°C) dry bulb and 61°F (16.1°C) wet bulb with a control range of not more than plus or minus 5°F (2.8°C).

3.9.1 Cooling capacity. A cooling load estimate shall be made and the system shall provide the conditions specified in 3.9, based on outside ambient conditions of 105°F (40.55°C) dry bulb and 78°F (25.55°C) wet bulb with 7.5 mile per hour winds. Internal heat gain from equipment shall be considered as 8 kW. Heat gain due to lights and evaporator fan shall be determined by the air conditioner or semitrailer manufacturer. The heat gain from occupants shall be based on 12 people sitting in the theater area, 2 people in the projection booth, and 2 people in the service area. In addition to the cooling capacity required to maintain the steady state design condition requirements, the cooling system shall be capable of cooling the semitrailer to 73.4°F (23°C) in approximately four hours after turning on the air conditioning system when the semitrailer has been heat-soaked for 24 hours at 105°F (40.55°C). Cooling load calculations shall be available to the engineering offices of the procuring activity.

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3.9.2 Heating capacity. A heating load estimate shall be made and the heating system shall provide the conditions specified in 3.9, based on outside ambient conditions of minus 20°F (minus 31.66°C) dry bulb with a 15 mile per hour wind. Internal heat gain from lights, indoor fan, equipment or other sources shall not be considered to offset heat loss. In addition to the heating capacity required to maintain the steady state design condition requirements, the heating units shall be capable of heating the semitrailer up to 73.4°F (23°C) in four hours after turning on the units when the semitrailer has been cold soaked for 24 hours at minus 20°F (minus 31.66°C). Heating load calculations shall be available to the engineering offices of the procuring activity.

* 3.9.3 System operation and controls. Components of the system and controls shall be designed for the conditions stated in 3.9, 3.9.1 and 3.9.2. A central control thermostat shall be provided to control the system. When two air conditioners or more than one heating unit are required, they shall be identical, with adjustable time delay relays in the compressor and heat control circuitry. Time delay relays shall be set in accordance with the manufacturer's recommendations to allow for staggered starting of the units. A control terminal box shall be mounted on the front interior partition and connected to the remote control thermostat. Parallel control cables shall be provided from the control terminal box to the air conditioning and heating units. Screw-on covers shall be provided for connectors. The entire electrical system for the air conditioning and heating units shall conform to 3.6.

* 3.9.4 Thermostat. The thermostat shall be designed to control the air conditioning and heating units. The thermostat shall be installed in the theater area, 4 to 5 feet above the floor, in a position to reflect the theater area temperature. The thermostat shall be tamperproof, requiring a key to change settings. The thermostat shall provide not less than the following features:

(a) System mode selector switches for:
    (1) Cooling (with the fan switch in the "On" or "Auto" position, the cooling cycle is energized and responds to thermostat requirements);
    (2) Heating (with the fan switch in the "On" or "Auto" position, the heating cycle is energized and responds to thermostat requirements);
    (3) Off (the air conditioner(s) are de-energized except that the crankcase heater(s) shall always be energized and the fan shall be controlled by a fan selector switch);
(b) Indoor fan selector switches for:
    (1) Auto (indoor fan operates in conjunction with heating or cooling system as required to satisfy the thermostat);

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(2) On (indoor fan runs continuously for ventilation when air conditioning system or mode selector is in "Off," or runs continuously when the system is in "Cool," or "Heat" mode);
(c) Temperature selector.
(d) Temperature indicator of actual temperature sensed.

* 3.9.5 Dampers (manual). Manually controlled dampers shall be provided in the ceiling for directing air flow from either air conditioner or any one heating unit, when only one is operational, through both ceiling ducts. Dampers shall be installed in the service area, in the theater area, and in the projection booth.

3.9.6 Cooling sequence. When the thermostat is set for the desired temperature, the fan set to "Auto," and the system mode selector set to "Cool," the cooling cycle shall be operational. When the conditioned space temperature rises above the thermostat set point such that the thermostat calls for cooling, the indoor or evaporator fan motor, the outdoor or condenser fan motor, other required controls, and compressor time delay relay shall be energized. The compressor time delay relay shall start the compressor approximately 15 seconds after the fans start. When conditioned space temperature satisfies the thermostat setting, the fan motors and compressor shall be de-energized. The compressor shall be locked out by a timer relay for approximately 5 minutes (or as the manufacturer recommends) to prevent short cycling.

* 3.9.7 Heating sequence. When the thermostat is set for the desired temperature, the fan set to "Auto," and the system mode selector set to "Heat," the heating cycle shall be operational. When the conditioned space temperature falls below the thermostat set point such that the thermostat calls for heating, the indoor fan motor, and the first stage time delay relay shall be energized. After the time delay relay times out, the first stage heating relay shall be energized and power the first stage of the electrical heating elements. If additional heat is required, the thermostat second stage heating contact shall close, energizing the second stage heat time delay relay number one. After this time delay relay times out, the second stage heating relay number one shall energize and power the second stage electrical heating element number one and energize the second stage heat time delay relay number two. When this time delay relay times out, the second stage heating relay number two shall be energized and power the second stage electrical heating elements number two. The first and second and all other electrical heat units shall cycle off and on as required to satisfy the thermostat.

3.9.8 Defrost cycle. The air conditioner(s) shall be equipped with an automatic defrost cycle. The defrost cycle may be controlled by either of several standard methods used by manufacturers, such as:
(1) A defrost pressure switch which closes on pressure drop through the

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outdoor coil and activates a timer motor. If the defrost thermostat contacts are closed (usually when the outdoor coil liquid line is at 45°F or below), the outdoor fan and reversing valve are de-energized, allowing the compressor to pump refrigerant through the coil to defrost it. This modified cooling cycle operates until either the defrost thermostat contacts open or until the defrost cycle timer switches back to heat cycle. Auxiliary electric heat is activated and de-activated when the defrost cycle is activated and de-activated; (2) The sequence is similar to that described in (1) above, except the pressure switch is omitted and the defrost cycle is activated by the defrost thermostat and a defrost timer is used to establish a time interval of 90 minutes from the end of a defrost cycle until the defrost thermostat can start another cycle.

3.9.9 Ventilation cycle. When the indoor fan selector switch is set to "On" and the mode selector switch is set to "Off," the system shall be de-energized and the fan shall run continuously.

3.9.10 Construction. The air conditioning and heating units shall be completely self-contained. The frameworks and casings shall be fabricated from aluminum or steel. The units shall be insulated and weatherproofed. The units shall be designed for mounting on the front of the semitrailer rearward of the double front doors and forward of the insulated partition in the service area. All areas of the units shall be accessible for service, replacement of parts, and general maintenance without injury to personnel and without removing the units from the vehicle. The units or all components thereof shall be mounted on shock absorbing cushions. Outdoor coil air intake and discharge openings shall be provided with protective grilles and with formed edge, channel shaped, rigid, hinged and gasketed door covers. Outdoor air intake gaskets and discharge door gaskets shall be of the smooth skinned or noncellular type. Doors and gaskets shall fit tightly to minimize road splash and dust entry during the transit mode. If the outdoor coil air intake or discharge openings are located on the side of the unit, provisions shall be made for the unit to be mounted and operated as either a right hand or left hand unit. Fresh air intake(s) with adjustable damper(s), cleanable filter(s), and rain deflector louvers shall be provided and shall be capable of being serviced and adjusted without the removal of the units. Conditioned air discharge ports shall mate with semitrailer air supply duct systems. Evaporator coil condensate drain pan, with piping through the bottom of the unit and a drain below the frame rails, shall be provided. Closure panels shall be provided for semitrailer openings when the units are removed.

3.9.11 Fans. The indoor fans shall be of the two-speed centrifugal type. The outdoor fans shall be of the single-speed, direct driven, propeller or centrifugal type. When fans are not direct drive type,

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double V-belts shall be used and means shall be provided for tension adjustment. Fan(s) shall be statically and dynamically balanced. Fan motor(s) shall be designed and rated for the temperature rise to be encountered. Motor(s) shall be sized for the actual total static pressure of the system and shall not overheat when operating at 120 percent of design conditions. When V-belt drive is used, sheaves and fan(s) shall be firmly keyed to shafts and secured with the proper type of set screws. End of shafts shall be threaded, convoluted, key-wayed and cross-drilled to accommodate a keyed thrust-washer and locknut. The thrust-washers and locknuts shall be parallel to the hub faces of the fan hub. Fan shaft bearings shall be selected to meet the design criteria, shall be of the permanently lubricated type, and shall have provisions for subsequent lubrication.

3.9.12 Piping, valves and accessories. Piping and tubing shall be copper, sized for proper flow rate without excessive pressure drop which might cause flashing of liquid refrigerant and cause faulty operation of the expansion valve. Piping and tubing shall be sized to insure oil entrainment required for lubrication of the compressor and the automatic valves. A strainer shall be installed ahead of each automatic valve. All components shall be located for ease of servicing and replacement.

* 3.9.13 Electric resistance heaters. Electric resistance heaters shall be installed. Heaters shall be operated with an air flow rate and sheath temperature as recommended by the heater manufacturer, but in no case shall air velocity over the face of the heater elements be less than 500 feet per minute nor sheath temperature exceed 800°F. Resistance heaters shall be fabricated from aluminized steel, nickel, or chrome steel alloys. Two high temperature limit switches (primary and supplementary) shall be provided and shall be UL rated for a minimum of 100,000 cycles of operation. The primary temperature limit switch shall be an automatic reset.

3.9.14 Air conditioning and heating. The air conditioning and heating units shall be located over the gooseneck at the forward end of the service area, and shall be accessible for service from the double front doors (see figure 7) and from the partition rearward of the air conditioners. The partition shall be insulated to prevent transfer of vibration and noise to the service area. Means shall be provided to remove, without damage, sufficient areas of the partition to perform any required service.

3.9.15 Ducts and diffusers. Delivery and return air ducts and diffusers shall be provided to suit the air conditioning and heating requirements specified herein. Ducts and diffusers shall pass all

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applicable fire safety tests for flame and smoke development. All duct work in the semitrailer shall be insulated. Diffusers shall be provided in sufficient quantity to meet the conditions imposed. Diffusers shall be of the adjustable type.

* 3.9.16 Air conditioner(s). The air conditioner(s) shall be sized to meet the performance requirements specified herein. However, in no case shall the air conditioner(s) be rated at less than 84,600 Btu/hour at 105°F dry bulb and 78°F wet bulb. Sensible heat ratio will be 0.81. Dehumidifying capacity shall be 1.8 gallons per hour. Fresh air shall be supplied at 280 cfm and shall be adjustable.

* 3.9.17 Heating. The heaters shall be sized to meet the performance requirements specified herein. However, in no case shall the heaters be less than 5 stages of 5 kW each.

3.9.18 Wiring. Units shall be wired from the main power distribution panel through rigid conduit above the ceiling and in the wall to a junction box. An electrical connector plug shall be provided for plugging in the units from inside the semitrailer. The connector plug shall be of the locking type, having suitable electrical characteristics. The systems shall conform to 3.6.

3.9.19 Treatment and painting of air conditioner(s) and heaters. Treatment and painting of the air conditioning and heating units and systems shall conform to that specified for the semitrailer exterior except for the following:

(a) Internal components, when painted by the component manufacturer, need not be repainted except when required to conform to (c) below;
(b) Refrigerant coils, piping, valves, electrical components, or other components whose function would be impaired by painting shall not be painted;
(c) Dissimilar metals shall be insulated from each other in accordance with MIL-STD-193;
(d) All surfaces which can be seen from the outside when the units are operating, with the front panels removed or the double front doors open, shall be painted white to match the exterior color specified.

* 3.9.20 Safety. The units shall have automatic safety controls which will shut down the units and protect against overheating and overcurrent.

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3.9.21 Human factors engineering. The units shall be engineered and designed in conformance to section IV of MIL-STD-1472.

3.10 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: focusing of lights; adjustment of electrical and braking systems; balancing of the suspension system; lubrication; actuation of the expansible sides and adjusting if required for proper operation; actuation of the leveling jacks; folding and unfolding of the floor; filling and charging of the batteries; wheel balancing; inflation of all tires; and complete lubrication of running gear and mounted equipment with grades of lubricants recommended for the ambient temperature at the delivery point.

3.11 Safety. All equipment or exposed portions of the equipment which create extreme temperatures or which could be damaged by inclement weather, and all rotating or reciprocating parts which are of a hazardous nature or located as to become a hazard to operating personnel, shall be insulated, fully enclosed, shielded or properly guarded.

* 3.12 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. All parts shall exhibit no evidence of damage, mutilation, or poor workmanship. Damaged components or parts that have been subjected to rebuild or repairs shall not be used as replacement parts or components. All mounting devices such as bolts, nuts and rivets shall be complete and secured in accordance with the highest standards of the industry. The overall appearance of the vehicle and all components thereof shall be pleasing and attractive.

* 3.12.1 Finish painting. All painted surfaces shall be free from bleeding, blistering, blushing, chalking, cracking and checking, crazing, dirt dullness, feather-edge splitting, fisheye, lifting, mottling, orange peel, peeling and chipping, pinholing, popping, runs and sags, sand scratches, shrinking, streaking, water spotting and wrinkling.

* 3.12.1.1 Definitions for painting.

a. Bleeding: showing through of original paint or primer into finish coat.
b. Blistering: bubbles in the finish.
c. Blushing: milky haze or mist that appears in a paint finish.
d. Chalking: lack of gloss and powdery surface.

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e. Cracking, checking and crazing: appearance of fine, irregular splits and cracks.
f. Feather-edge splitting: separating or cracking of the finish coat.
g. Fisheye: holes in the paint film which allow the underlying finish to be seen.
h. Lifting: shriveling of the paint surface.
i. Mottling: spotty or streaked appearance (Metallic paint).
j. Orange peel: bumpy surface (much like the skin of an orange).
k. Pinholing: tiny holes in the finish.
l. Popping: blistered appearance.
m. Runs and sags: overspray conditions.
n. Sand scratches: scratches showing through the finish coat.
o. Shrinking: paint that separates from area repaired with plastic filler.
p. Streaking: zebra-like effect.

* 3.12.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not degrade the design strength of the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

* 3.12.3 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts and screws shall be tight.

* 3.12.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of an approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and shall be in full contact with the surface of the member.

* 3.12.5 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

* 3.12.6 Paneling installation. All interior paneling shall be installed uniformly, free of misalinement, open seams, gaps, and uneven surfaces. Interior paneling shall be properly secured, free of

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misalined joiners, edgings, waviness, dents, nicks, sharp edges, and other forms of deformation or surface defects. Exterior paneling shall be free of damaged or missing rivets. Exterior overlapped sheet surfaces sealed with sealing compound shall show sealant in uniform application.

* 3.12.7 Door installation. All doors shall be properly located, fitted and sealed to prevent entrance of noise, air, dust and water. All doors and associated hardware shall operate without binding.

* 3.12.8 Installation of insulation. The insulation shall be installed in a manner excluding excessive voids, open spaces, rips, tears and other types of damage or defects. The insulation shall be installed and secured in a manner to retain the material in a permanent position under all operating conditions.

* 3.12.9 Floor installation. Plywood used to enclose the floor area shall be securely installed and free of gaps, open seams, cuts, or open holes. All wood parts shall be free of cracks or splinters.

* 3.12.10 Electrical system installation. The electrical systems shall be installed to provide proper routing for electrical wiring with clearances between wiring and applicable parts. All wiring shall be secured and fastened at all points of bending and protected against chafing and damage. Wiring shall be concealed and easily accessible for repairs. All circuits shall be properly grounded. Electrical wiring covering shall be free of cracks, cuts, kinks, frays, or similar defects. All lighting, wiring, switches and receptacles shall be securely installed, straight and true, and free of loose or faulty connections. Wall outlets and other receptacles shall be conveniently located and installed uniformly. All electrical equipment and fixtures shall be mounted securely and properly positioned to prevent damage during transit.

* 3.12.11 Air conditioning and heating installation. Air conditioning units shall be securely mounted. The units shall be properly installed and fitted. The air conditioning units shall perform satisfactorily without vibration or undue noise. The heaters shall be mounted securely. The heater installations shall be free from gaps, exposed electrical wiring or exposed insulation. The fan blowers shall operate free from rattles, vibration and any other undue noises or malfunctions.

* 3.12.12 Ceiling surfaces. All ceiling surfaces shall be securely installed, properly fitted, alined and leveled. Surfaces shall be free of scratches, mars, uneven surfaces, rough or chipped edges or corners, and color fade.

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* 3.12.13 Stairways and handrails. The stairways and handrails shall be of rigid construction and securely mounted. The surfaces shall be free of burrs, nicks and scratches and the handrails shall be smooth.

* 3.12.14 Fan installation. Fans shall be securely mounted and fitted, free of gaps, openings, exposed electrical wiring or exposed insulation. The installation shall be rainproof and waterproof. The exterior sealant shall be uniform, neat in appearance and free of bare or thin areas. The fans shall operate without rattles, undue noises or malfunctions.

* 3.12.15 Plates and boxes. Data plates and electrical receptacle and switch cover plates and boxes shall be installed in a neat workmanlike manner and shall be straight and true.

* 3.12.16 Rustproofing application. Readily visible areas shall be completely cleaned and free of rustproofing compound (see 3.1.1.6).

* 3.12.17 Visible seals. Visible seals shall be straight and true and shall not detract from the appearance of the vehicle.

* 3.12.18 Sealer. All visible sealer not required for proper sealing shall be removed.

* 3.12.19 Trim. Workmanship shall display no sharp edges, dents, gaps or spaces between pieces of trim. Corners shall be mitered.

* 3.12.20 Locks and handles. Door locks and handles shall be attractive in appearance, straight and true.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection

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system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility for assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

* 4.3 First vehicle preliminary inspection. During fabrication of the first vehicle produced under the contract, an in-process inspection will be conducted by representatives of the Army Tank-Automotive Command (TACOM) to evaluate conformance and workmanship. The examinations and tests subject to the in-process inspection, as identified below, shall be accomplished as specified with appropriate advance notification to the Government, as stated in the contract (normally fourteen days prior to the production stage to be inspected).

* 4.3.1 Waterproofness test. The semitrailer shall be subjected to the following water tests conforming to 3.3.5. All water tests shall be conducted utilizing a rigid fixture framework capable of maintaining constant pressure and consistent direction during the entire test.

* 4.3.1.1 Skin and seal water test. The in-process skin and seal water test shall be accomplished prior to installation of roof and wall insulation, interior finish, etc., which might obstruct or absorb any notable weepage. The extended side sections shall have a payload and a compensatory load placed within the expansible section to simulate a loaded, completed interior. Weepage that exceeds a tear drop in a given section or a tear drop in more than three sections shall be cause for rejection.

* 4.3.1.2 Service life water test. After completion of the service life test, a second water test shall be accomplished. The seals shall maintain waterproofness integrity as limited in 4.3.1.1.

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* 4.3.2 Service life test. The contractor shall perform a preliminary in-process inspection to demonstrate compliance to 3.3.10. The performance test shall consist of fully expanding and retracting the sides for 10 sets of 35 cycles or a total of 350 cycles. The seals shall be examined for wear and tear and the unit shall pass a waterproofness test. The seals and any worn or fatigued parts shall be replaced prior to acceptance.

* 4.4 First vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract utilizing appropriate examinations, tests and inspections to demonstrate that all provisions and requirements have been adequately addressed. Aside from the performance of the basic inspection, the contractor shall have any supporting objective evidence readily available for review by the Government. The objective evidence shall include, but not be limited to, such items as purchase orders, certifications, test reports, in-process inspection records, specimens and similar data which demonstrate conformance to the contract. Acceptance of the first vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.4.1 Curb weight distribution. The vehicle shall be weighed to determine the curb weight, at the ground, on the rear tandem axles, the landing gear and the kingpin as specified in 3.2.3.

* 4.4.2 Treatment, painting, marking and instruction plates. Treatment, painting, marking and instruction plates shall be in accordance with 3.1.1.1 through 3.1.1.2.3, except that the first vehicle may be evaluated in the prime paint until completion of all tests. The vehicle shall then be finish painted prior to submission for final inspection.

* 4.4.3 Road test. The road test shall be done with a full payload. This test shall include the following:

(a) At least 150 miles over primary and secondary paved, hard surface roads having random undulations and chuck holes at varying speeds up to 55 mph. The last 50 miles of this test will be accomplished at sustained speeds of at least 55 mph;
(b) At least 100 miles over unpaved, secondary roads at speeds up to 35 mph;
(c) Five sudden stops on roadways described in (a) above at speeds of 15, 25, 35, 45 and 55 mph;
(d) Five sudden stops on roadways described in (b) above at speeds of 15, 20, 25, 30 and 35 mph;

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(e) Upon completion of the road tests, the vehicle and components shall be operable in the normal manner and shall show no evidence of malfunction or failure. The road test vehicles shall be refurbished prior to being offered for final acceptance.

* 4.4.3.1 Emergency brake. The emergency brake shall be capable of holding the semitrailer with a full rated payload on a 10 percent grade while resting on the landing gear, detached from the towing vehicle. All air shall be drained from the tanks and there shall be no detectable movement during the 30-minute duration of the test.

* 4.4.3.2 Breakaway test. A breakaway test shall be performed by removing the truck tractor's air supply lines from the semitrailer, then attempting to tow the semitrailer. The wheels of the semitrailer shall either be locked, or significant resistance shall be evidenced in repeated attempts to tow the semitrailer for not less than 15 minutes.

* 4.4.3.3 Inspection after testing. Upon completion of the tests, the vehicle shall be thoroughly inspected for evidence of damage, warpage, breakage, loose connections, popped rivets or screw heads, etc. The semitrailer body and body mountings shall be visually inspected to assure that the following areas do not indicate any manfunction or deformation. The following checks shall be performed:

(a) Extend and retract the body sides, both mechanically and by use of the electric motor.
(b) Fold and unfold the floor.
(c) Operate the landing gear, both manually and with air power.
(d) Open, close, lock and unlock all doors.
(e) Examine stowage compartment interior and mountings.
(f) Examine interior lighting and lenses.
(g) Examine air conditioner and heater mountings.
(h) Examine all visible electrical circuits.
(i) Functionally test all components and fixtures, including assembly, knockdown and stowage of the stairways and handrails.
(j) Electrical test to determine that all vehicle electrical systems and fixtures are functional and capable of operating at the required capacities and modes. The test shall be done with actual or simulated loads without incurring any excessive heat or power interruption for a 30-minute period. Electrical power shall be supplied from the engine-generator on the truck tractor.

* 4.4.3.4 Final waterproofness test. Upon successful completion of the road test and loading, the vehicle shall receive and pass a final waterproofness test.

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* 4.4.3.5 Lubrication and seal replacement test. Subsequent to 4.4.3.4, the lubrication time and the seal replacement time shall be demonstrated to be in conformance to 3.3.11 and 3.3.12, respectively.

4.4.4 Workmanship. All workmanship and manufacturing techniques shall be of the highest commercial standards. All fabrication steps shall result in exacting fits free of burrs, chips, kinks and similar defects. Reworked items shall be acceptable only if they result in like-original-condition parts and fit. The final finish process shall be inspected by the contractor during sanding, priming, painting and repainting to assure that the entire procedure culminates in an automotive type, high quality finish. Proper masking, pre-cleaning and preparation shall be such as to preclude overspray on unpainted surfaces such as mouldings, trim and accessories. The final paint coat of the semitrailer and the truck tractor shall be an exact match as to paint color cited, to include any degradation caused by time, technique, brand or source. The entire vehicle shall be critically inspected to determine compliance to 3.12 through 3.12.20. The entire package, when presented, shall be of showpiece caliber and reflective of the image and pride of the US Armed Services.

* 4.5 Production sample. Upon acceptance of the first vehicle, it shall remain at the manufacturing facility as a production standard and shall be the last vehicle shipped on the contract. The contractor shall be responsible for assuring that the production sample is maintained and leaves the plant in acceptance condition.

* 4.6 Failure. Failure of the first vehicle to meet the requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

* 4.7 Inspection of production vehicles. Each production vehicle (less payload) shall be examined and driven on an approximately level road surface by the contractor to assure that the vehicle will operate in accordance with the contractual requirements. Each production vehicle shall be subjected to a water test with the same spray and pressure cited in 3.3.5. The water shall be directed at all four sides, at the roof and at the underbody for 15 minutes in the retracted position and for 15 minutes in the expanded position. The vehicle shall show no signs of water penetration as limited by 4.3.1.1. Each vehicle shall be functionally tested in accordance with 4.4.3.3.

5. PACKAGING

* 5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

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6. NOTES

* 6.1 Intended use. The vehicle covered by this specification is intended for nontactical use as a mobile recruiting vehicle.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.

* 6.2.1 Towing vehicle. The towing vehicle will be a diesel engine driven, 6-wheel, 4-wheel drive truck tractor with an automatic five-speed transmission, and a fore and aft rocking, sliding fifth wheel. The truck tractor will be equipped with a diesel engine driven generator having a rating of not less than 40 kW. The generator will provide 120/240 volts, three phase, 60 Hz, 5-wire. In addition, the generator will accommodate a 120-volt single phase cable when 240 volts is not required.

* 6.3 Subject term (key word) listing.
Air conditioning system
Air fifth wheel
Air suspension
Electrical system, 120/240-volt
Expansible sides
Heating system
Miter boxes
Recruiting vehicle
Semitrailer, van, expansible.

* 6.4 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the previous issue.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2330-A958

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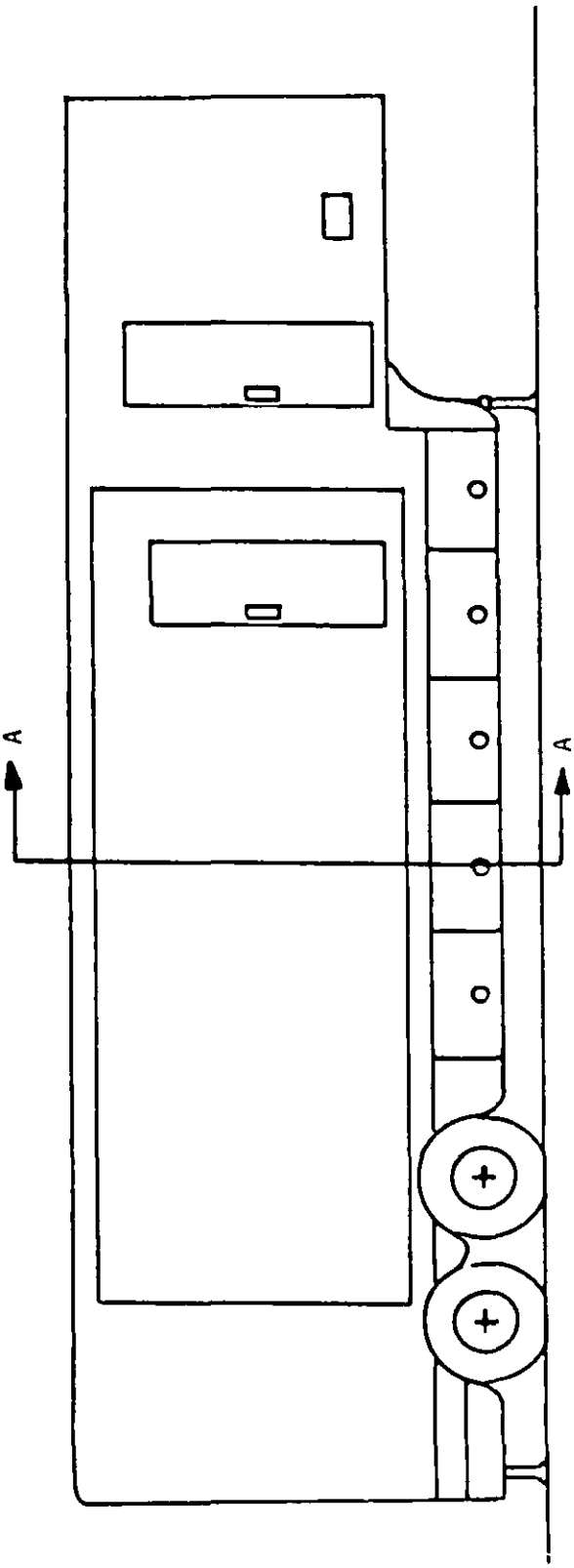


FIGURE 1. Semitrailer, van, expansible; curbside view.

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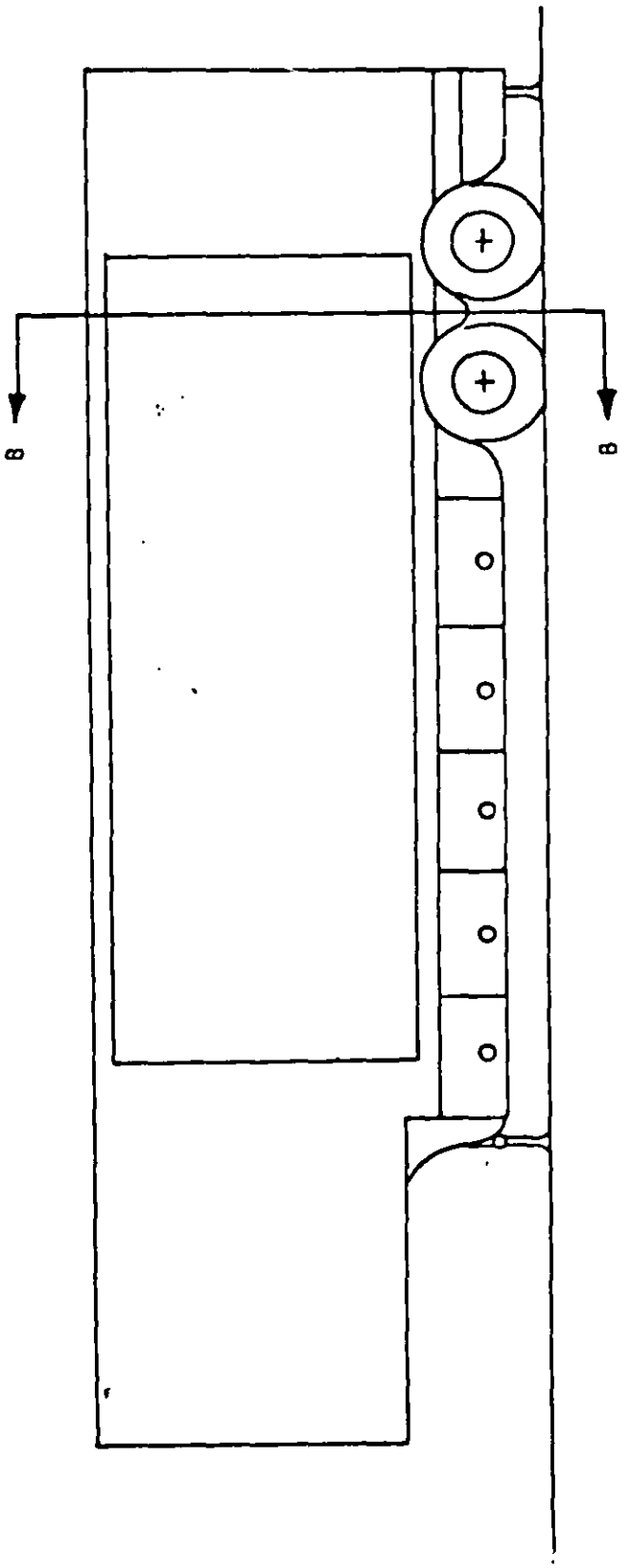


FIGURE 2. Semitrailer, van, expansible; streetside view.

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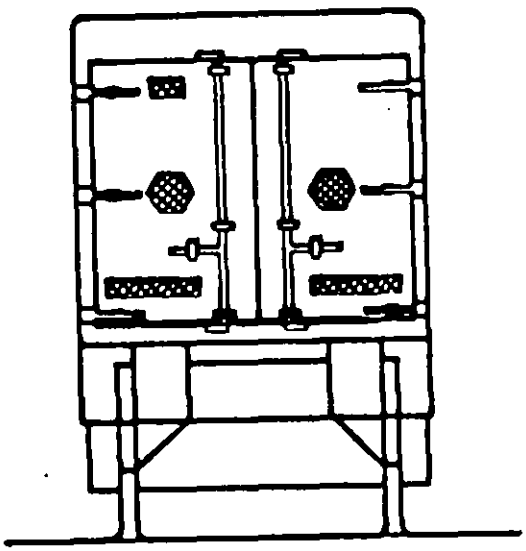


FIGURE 3. Semitrailer, expansible, sides retracted; front view.

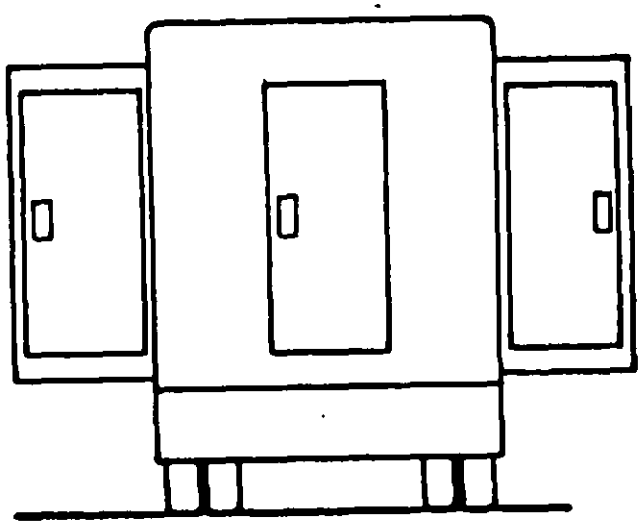
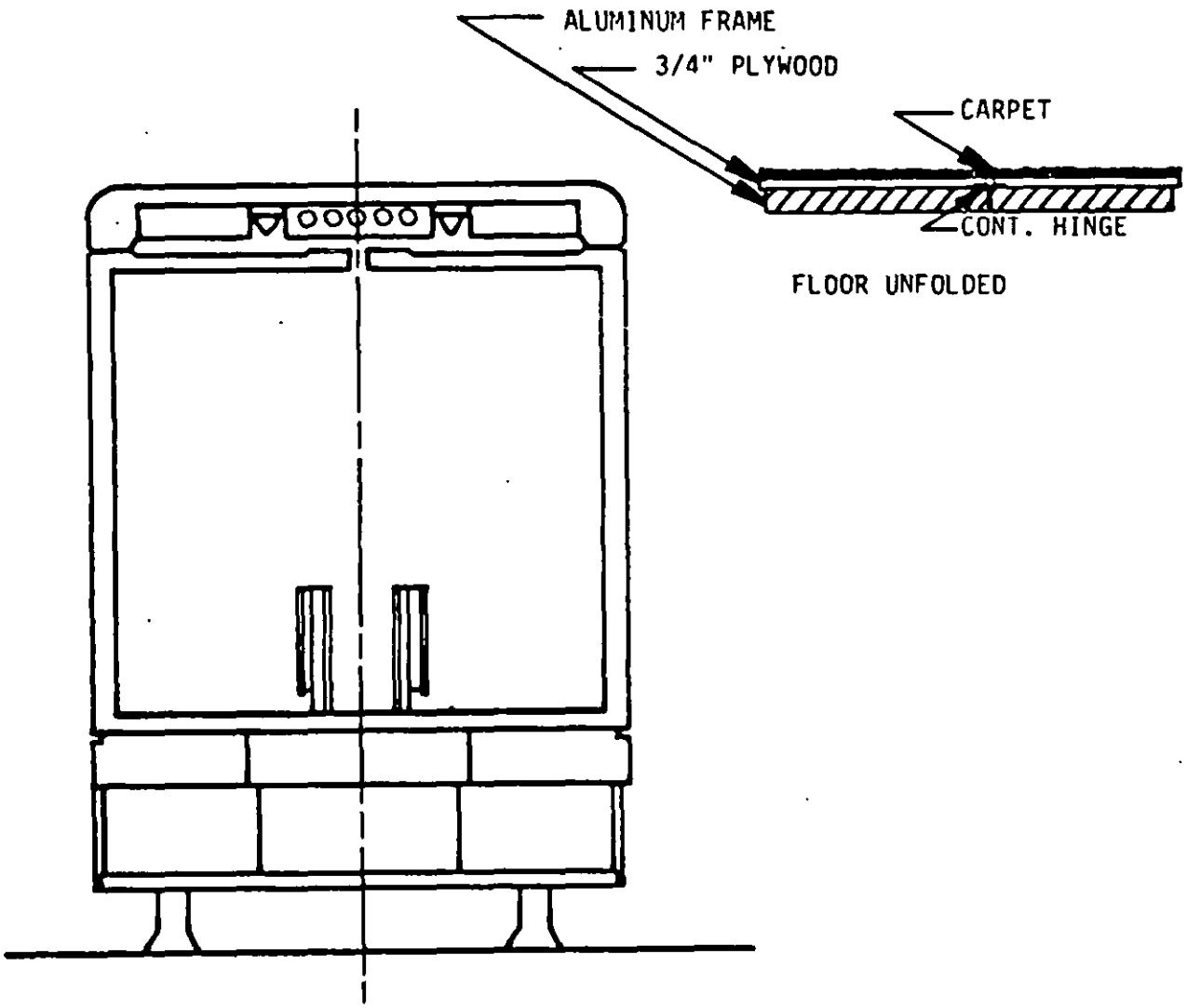


FIGURE 4. Semitrailer, expansible, sides expanded; rear view.

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REF: SECTION AA. FIGURE 1

FIGURE 5. View through semitrailer, retracted (theater area).

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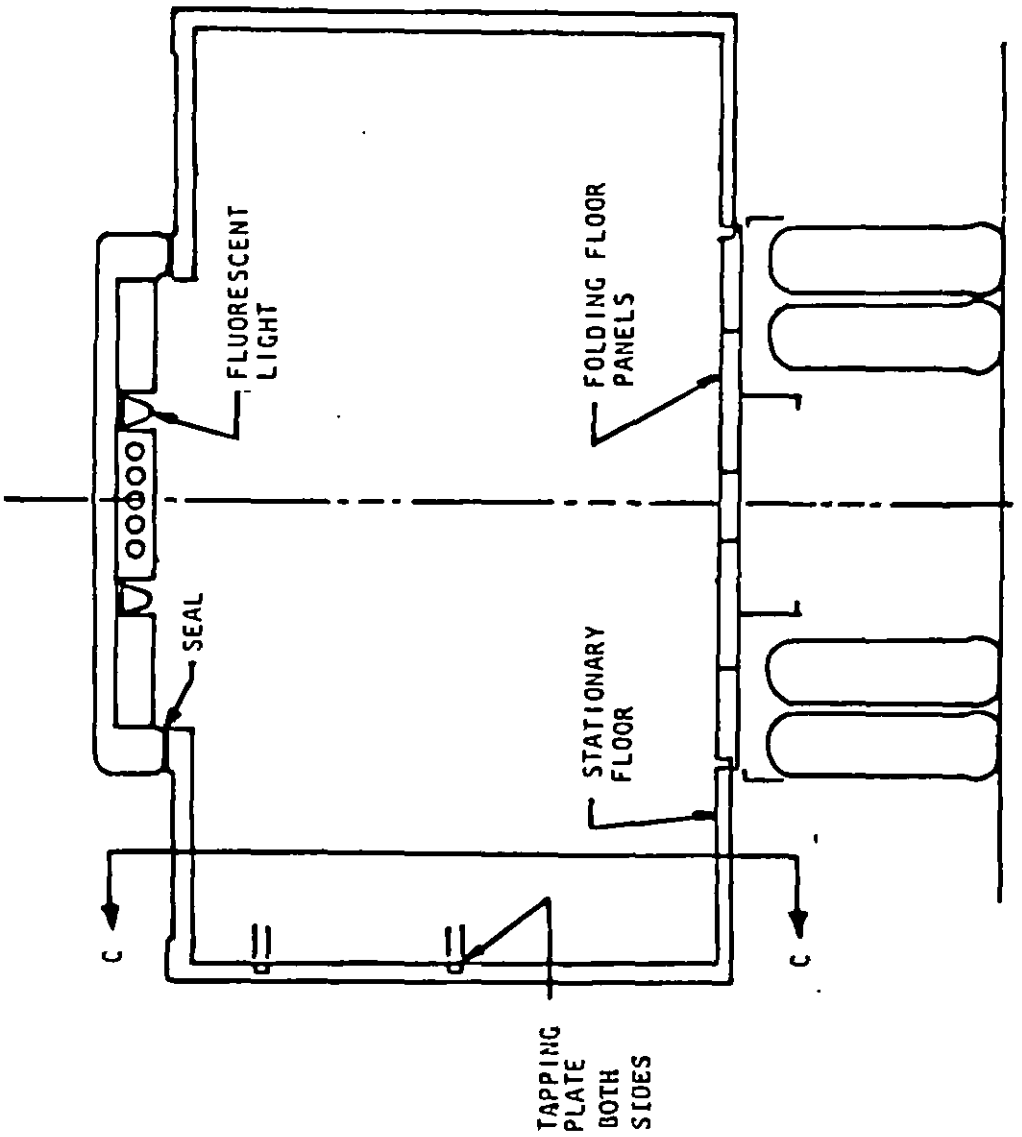


FIGURE 6. View through semitrailer, expanded (theater area).

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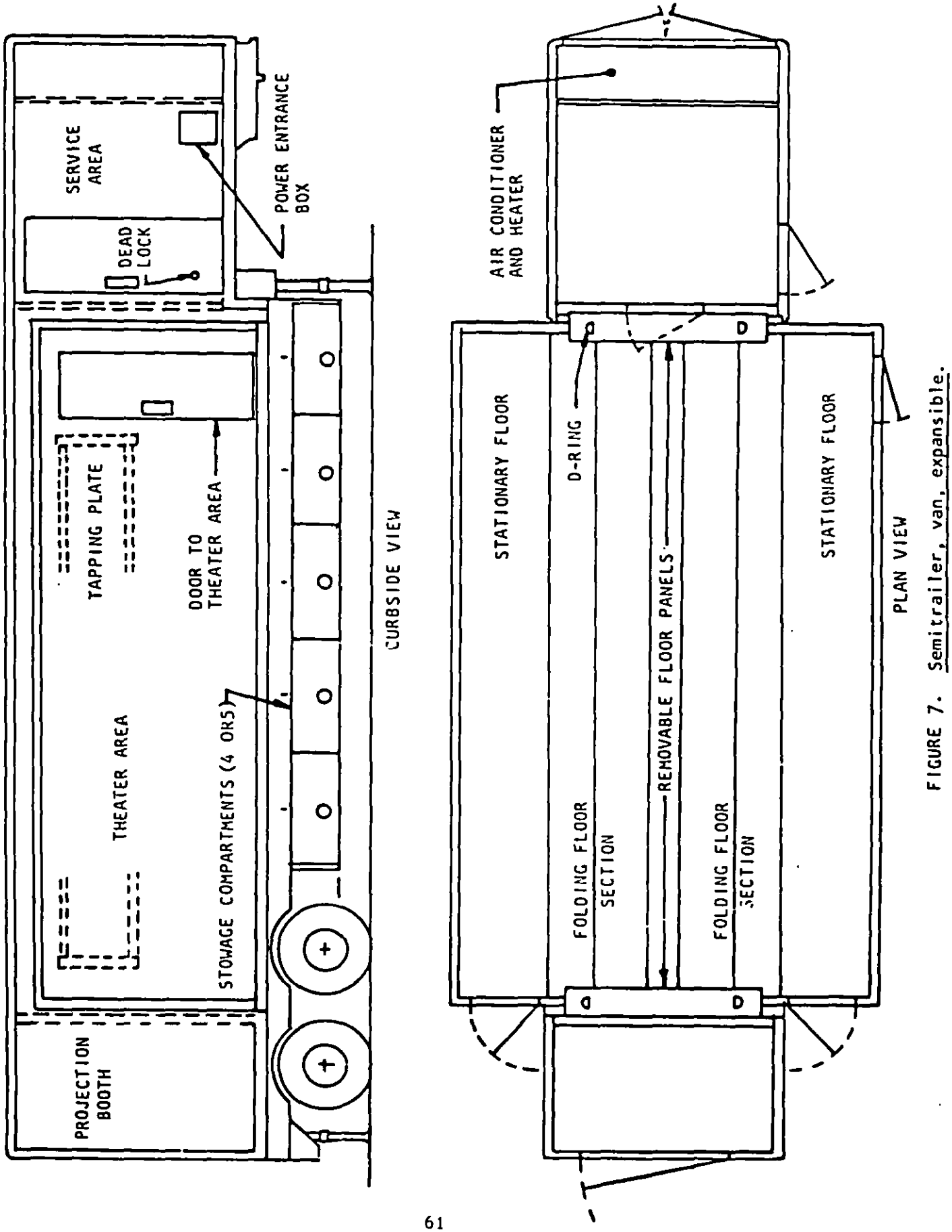


FIGURE 7. Semitrailer, van, expansible.

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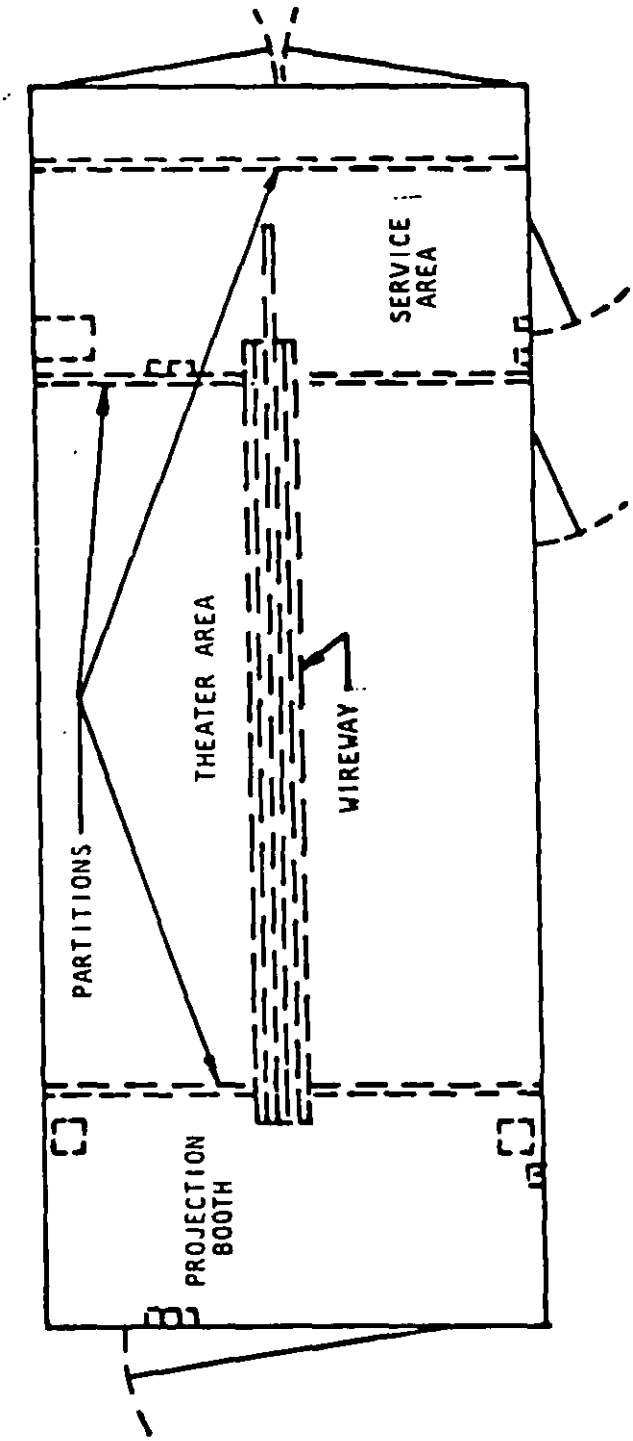


FIGURE 8. Plan view of partitions and wireway (sides retracted).

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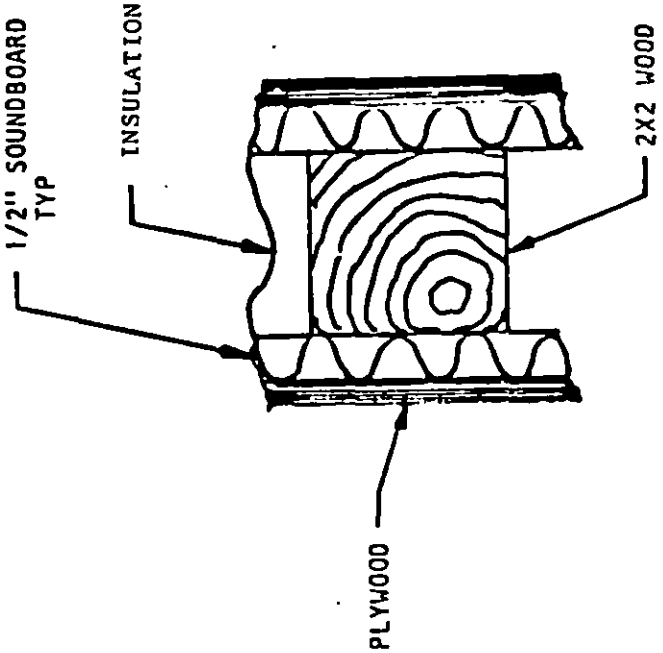


FIGURE 9. Section through rear partition.

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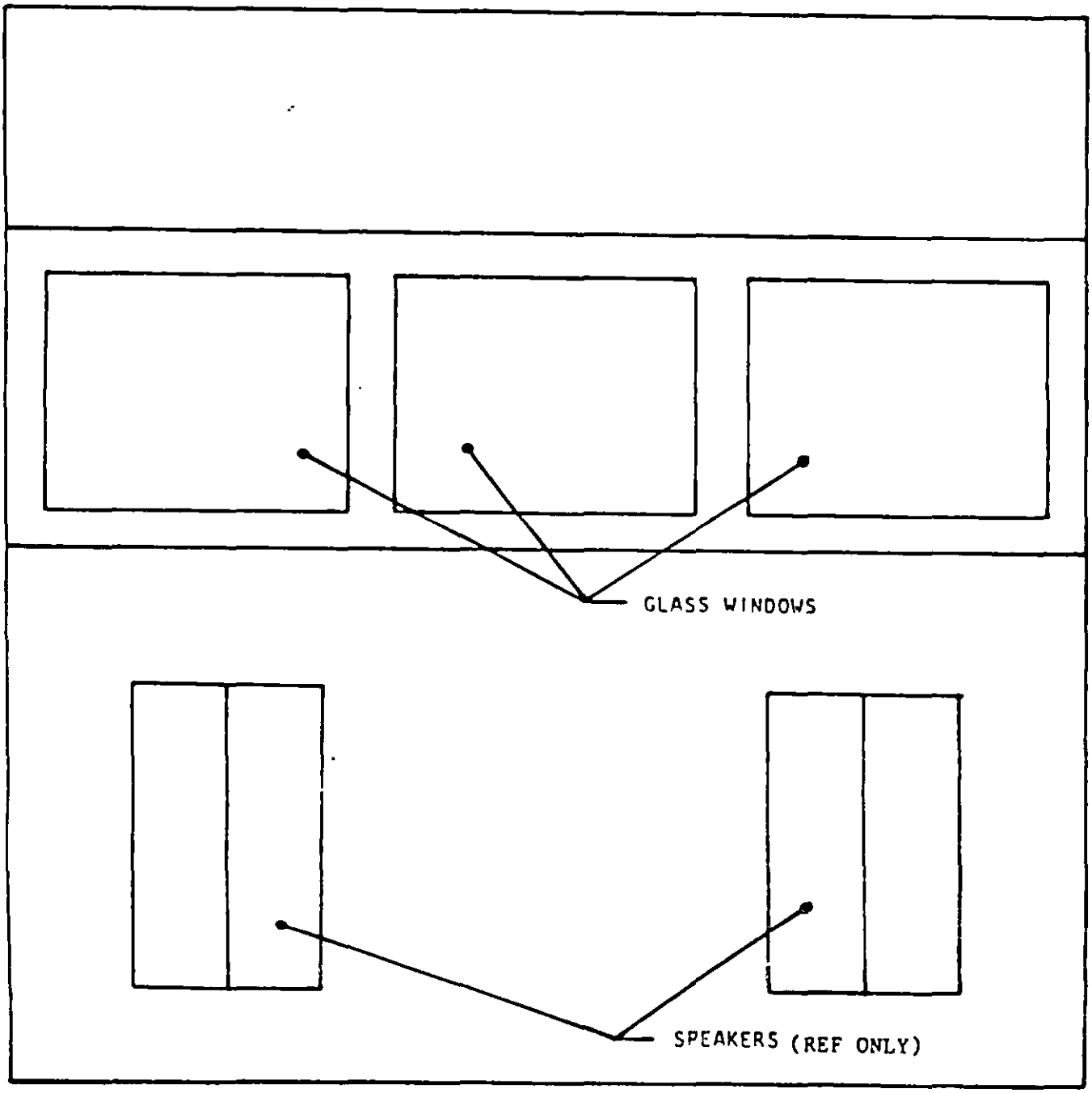


FIGURE 10. Projection booth wall, view from theater area.

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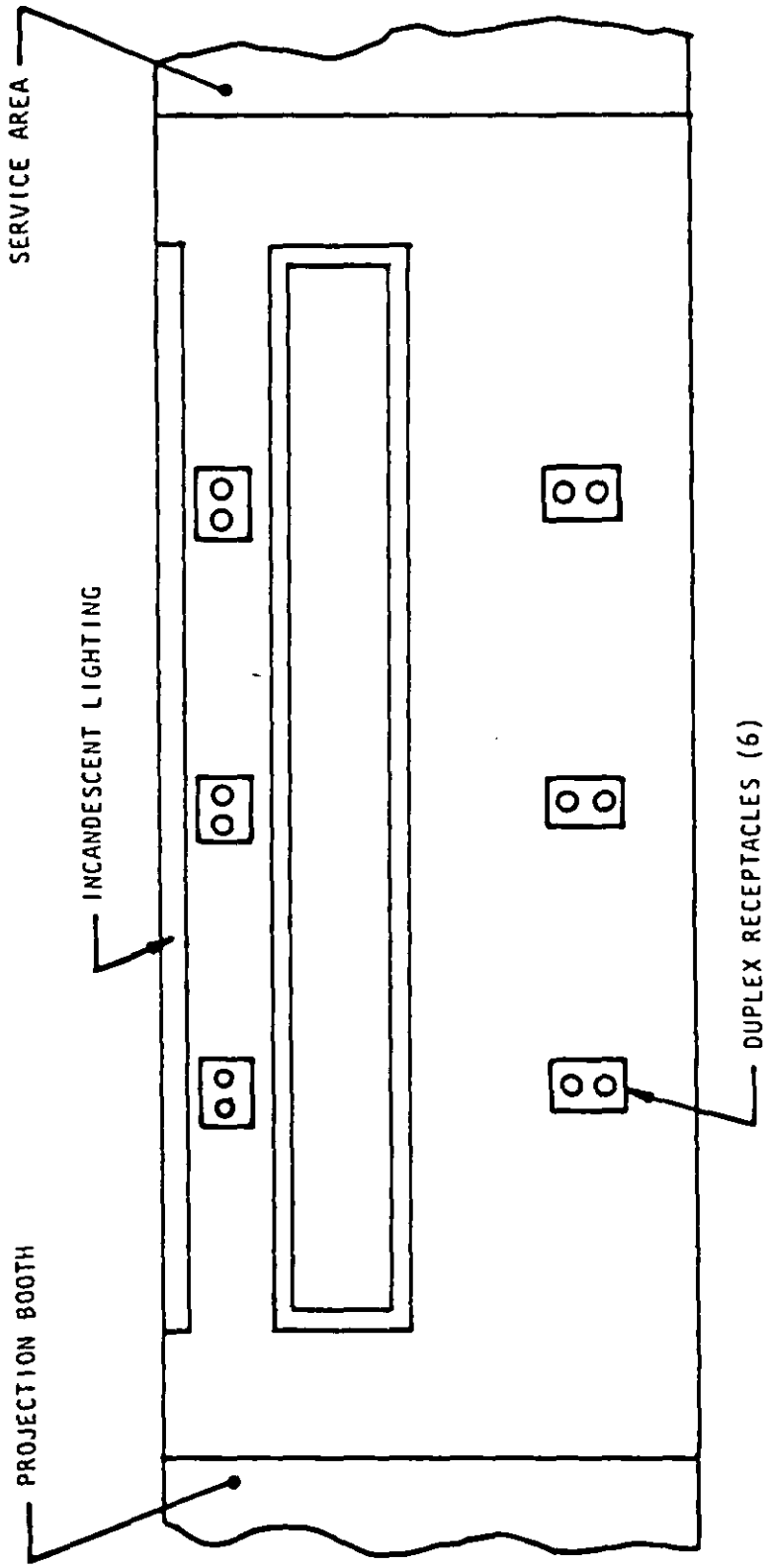
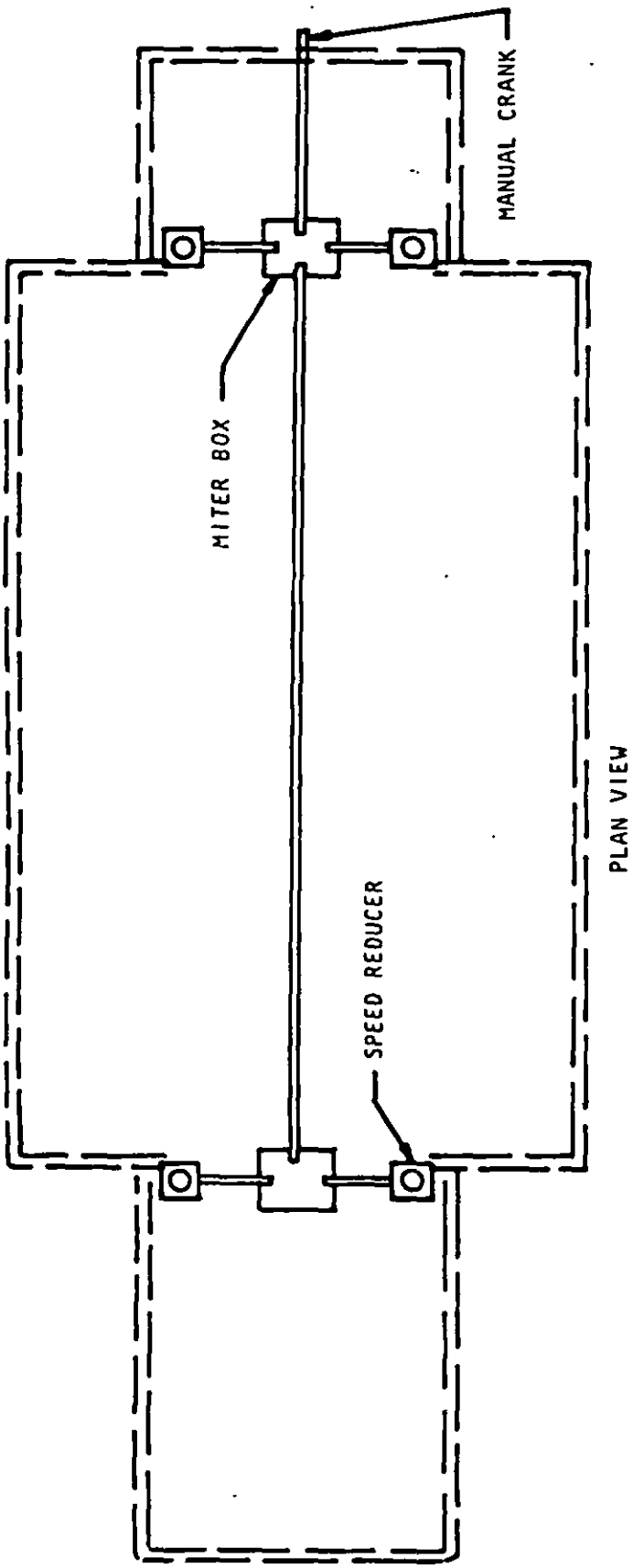


FIGURE 11. Interior view of expansible wall - streetside.

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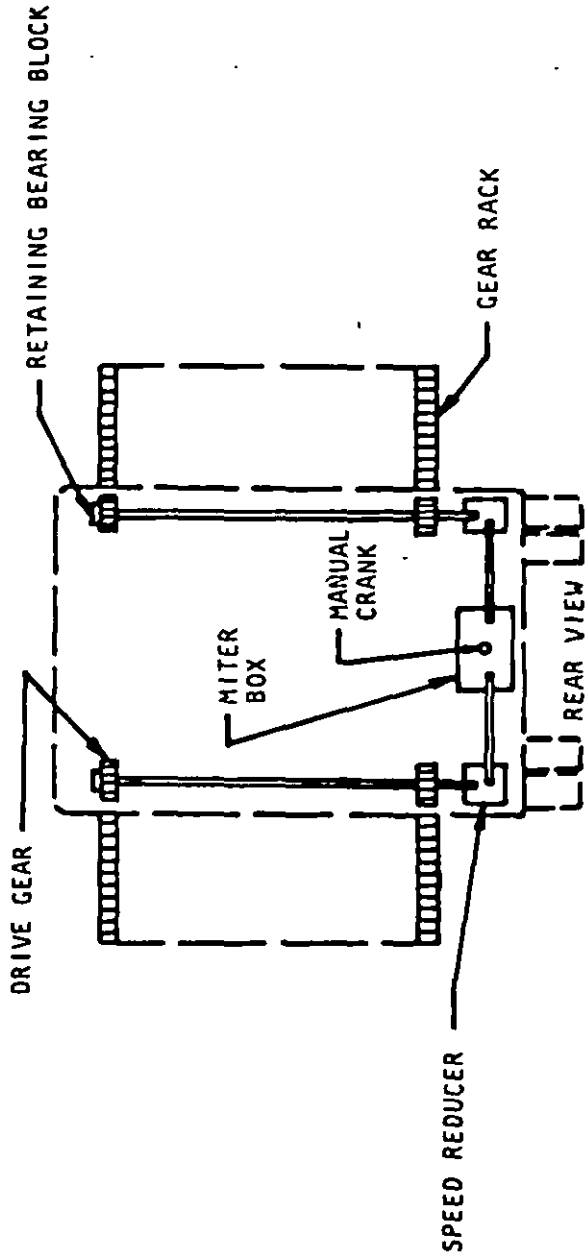


FIGURE 12. Power transmission.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-62472A	SEMITRAILER, VAN: EXPANSIBLE, EXHIBITION, 12-TON COMMERCIAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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